UNIVERSITY OF ECONOMICS AND LAW
FACULTY OF INFORMATION SYSTEMS

PROJECT REPORT
BUSINESS INTELLIGENCE & DECISION SUPPORT SYSTEMS

TOPIC: DEVELOPING BUSINESS INTELLIGENCE (BI) SOLUTIONS IN ADVENTURE WORK CYCLES COMPANY

Group: 13

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Ho Chi Minh City, July 2025

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ACKNOWLEDGMENT

Throughout the course of completing our capstone project for the Business Intelligence and Decision Support Systems module, we have received valuable support from numerous individuals and organizations. On the occasion of finalizing this report, we would like to express our sincere gratitude to all those who have accompanied, guided, and provided favorable conditions for us during our learning and research journey.

First and foremost, we would like to extend our heartfelt thanks to Mr. Lê Bá Thiền, M.Sc., the course instructor, who has enthusiastically imparted both foundational and in-depth knowledge in the field of Business Intelligence. His dedicated teaching, continuous guidance, and insightful direction have played a vital role in helping us complete this project.

We would also like to express our appreciation to the University of Economics and Law – Vietnam National University, Ho Chi Minh City, especially the Faculty of Management Information Systems, for providing us with the necessary facilities, materials, and a conducive academic environment to successfully complete the course.

In addition, we are grateful to our senior peers and fellow students, who have generously shared their experiences, provided feedback, and supported us with technical assistance and reference materials throughout the project implementation process.

Despite our utmost efforts, we acknowledge that this report may still contain certain limitations. We respectfully welcome constructive feedback from our esteemed lecturers to further improve the quality of our future research and academic endeavors.

Once again, we sincerely thank you all.

COMMITMENT

We hereby declare that this capstone project for the Business Intelligence and Decision Support Systems course is the result of our group's serious and diligent learning, research, and collaboration. All content, data, images, tables, and analytical results presented in this report were either conducted directly by our team or properly cited in accordance with academic regulations.

We sincerely appreciate the attention and evaluation of our respected lecturers.

Ho Chi Minh City, July 1st, 2025

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ACRONYMS

Acronyms	**Meanings**
BI	Business Intelligent
OLAP	Online Analysis Processing
KPI	Key Performance Indicator
SCD	Slowly changing dimension
ETL	Extract - Transform - Loading
OLTP	On-line transactional processing,
SSIS	SQL Server Integration Services
SSAS	

CHAPTER 1: INTRODUCTION

1.1 Business case for the project

1.1.1 Business case overview

1.1.1.1. Market Analysis

a. Global Market Trends

The worldwide bicycle sector has demonstrated robust growth, with forecasts pointing to continued expansion. Per Statista (2025), the global bicycle market is projected to achieve a revenue of $63.73 billion in 2025. Analysts anticipate a compound annual growth rate (CAGR 2025–2029) of 2.37%, culminating in an estimated market value of $69.99 billion by 2029. By the same year, global unit sales are expected to total 138.29 million bicycles, with the average price per unit in 2025 forecasted at $463.12. Internationally, China is set to dominate revenue generation, contributing an estimated $12 billion in 2025 (Statista, 2025).

The market's growth is propelled by the rising adoption of electric bicycles (e-bikes) and the increasing appeal of cycling as an eco-conscious and leisure activity. E-bikes, combining pedal power with electric assistance, are gaining traction for their environmental benefits and ease of use, particularly for extended travel or rugged landscapes. Additionally, demand is surging for purpose-built bicycles, such as those for road racing or mountain biking, fostering the growth of specialized manufacturers. Regional factors, including urban congestion and vibrant cycling communities, significantly influence market trends. Moreover, broader economic factors, such as government support for green transportation and a global focus on fitness and sustainability, are driving widespread bicycle use.

b. Vietnam Market

The bicycle industry in Vietnam is experiencing a steady upward trajectory, though its scale remains modest in comparison to global leaders. According to Statista (2025), the Vietnamese bicycle market is projected to generate revenues of approximately $98.80 million in 2025. Forecasts indicate a compound annual growth rate (CAGR) of 2.35% from 2025 to 2029, culminating in an estimated market volume of $108.43 million by 2029. Unit sales are expected to reach 393,640 bicycles by 2029, with an average weighted price of $235.92 per unit in 2025 (Statista, 2025). In contrast, China

is anticipated to lead the global market, generating an estimated $12 billion in revenue in 2025, highlighting the relatively smaller scope of Vietnam's market, which is shaped by economic conditions and limited public awareness of cycling as a sustainable transport option.

This growth is driven by several key factors, including rapid urbanization, increasing environmental awareness, and a rising demand for sustainable transportation solutions. Electric bicycles (e-bikes), which offer an eco-friendly alternative with reduced physical exertion, are gaining significant traction, particularly for urban commuting. The e-bike segment is poised for substantial growth through 2029, reflecting evolving consumer preferences. Concurrently, demand for specialized bicycles, such as those designed for road cycling or mountain biking, is fostering the emergence of niche manufacturers. Local conditions, including urban congestion and the gradual development of cycling communities, further influence market dynamics. To fully harness this potential, strategic investments in cycling infrastructure, such as dedicated bike lanes, and initiatives to promote cycling's health and environmental benefits are essential. For example, in January 2024, the Hanoi Department of Transport announced a 4km bicycle-only lane along the To Lich River, a development expected to bolster cycling adoption (IMARC Group, 2024). Such measures are critical to strengthening Vietnam's bicycle market and advancing its role in the nation's sustainable transportation framework.

1.1.1.2. Company overview

Adventure Works Cycles, a fictional multinational bicycle manufacturer featured in Microsoft Learn resources, is headquartered in Bothell, Washington, and employs approximately 300 dedicated professionals. Renowned for its commitment to innovation and market expansion, the company has strategically enhanced its operations to strengthen its global presence. In 2000, Adventure Works Cycles expanded its production capabilities by acquiring the Importadores Neptuno facility in Mexico. This plant is integral to the company's operations, producing essential bicycle components and serving as the exclusive manufacturing hub for its touring bicycle line.

To bolster its market position, Adventure Works Cycles is investing in a robust digital platform to deliver comprehensive product information and enhance customer accessibility. Simultaneously, the company focuses on optimizing production processes to reduce costs, thereby maximizing profitability while maintaining high-quality offerings for its customers.

1.1.1.3. Role of the Purchasing Department

The purchasing department, often referred to as the procurement or acquisition division, is a cornerstone of organizational efficiency, as outlined by industry insights from Indeed Editorial Team. This department is tasked with securing the goods and services necessary for seamless business operations, contributing to both daily functionality and long-term strategic objectives. The scope of its responsibilities varies by organization size but typically includes overseeing supply chain operations, managing vendor relationships, and ensuring operational efficiency through effective contract management.

Functions of the Purchasing Department

The purchasing department plays a critical role in optimizing resource acquisition to support organizational goals. Its primary objectives include sourcing reliable suppliers, negotiating cost-effective agreements, and adhering to budgetary constraints. The department fosters strong supplier partnerships, secures favorable contract terms, and ensures timely delivery of goods that meet quality standards.

Beyond procurement, the department strategically manages inventory to minimize costs while ensuring adequate stock levels, requiring meticulous planning. It also mitigates risks by addressing potential disruptions, price volatility, and quality concerns within the supply chain. Compliance with legal, regulatory, and organizational standards is paramount, ensuring ethical practices are upheld.

Strategically, the purchasing department aligns its activities with the organization's long-term vision, monitoring market trends and providing valuable insights for strategic planning. By achieving these goals, the department enhances resource efficiency, reduces operational risks, and strengthens the organization's competitive position in a dynamic market environment.

1.1.2 Business demands/problems

1.1.2.1. SWOT analysis of the bicycle industry in the global market

Table 1-1. SWOT Analysis of the Global Bicycle Industry (Source: Authors' Proposal)

Strengths	Weaknesses
Eco-Conscious Transport: Bicycles are valued as a sustainable mobility option, resonating with global environmental priorities and driving demand for green transport solutions.	**Seasonal Market Variability**: Sales in the bicycle industry often fluctuate due to seasonal weather changes, with declines in colder or rainy regions affecting revenue.
Fitness and Lifestyle Appeal: The growing focus on health and wellness fuels demand for bicycles and related accessories, appealing to consumers prioritizing active lifestyles.	**Safety Risks**: Inadequate cycling infrastructure and concerns about road safety in urban areas may discourage potential users, limiting market expansion.
Urban Mobility Advantage: Rising urbanization encourages cycling as a practical, cost-effective solution for short commutes, reducing urban congestion and pollution.	**Competing Mobility Options**: Alternative transport modes, such as electric scooters and ride-hailing services, challenge bicycle sales, particularly for urban travel.
Technological Innovation: Advances in bicycle materials, design, and technology create high-performance products that attract consumers seeking innovative features.	**Cost Barriers**: The high cost of premium bicycles and specialized equipment can deter price-sensitive consumers, restricting market reach.

Policy Support: Governments worldwide promote cycling through infrastructure investments, bike-sharing initiatives, and commuter incentives, fostering market growth. For example, Copenhagen, Denmark, has completed four dedicated bicycle bridges.	**Market Fragmentation**: The presence of numerous small and medium-sized manufacturers creates a highly competitive market, leading to pricing pressures and reduced margins.

Opportunities	Threats
Growth in Emerging Markets: Rapid urbanization, rising incomes, and infrastructure development in regions like Asia, Latin America, and Africa present significant potential for bicycle market expansion.	**Regulatory Shifts**: Evolving regulations on safety standards, tariffs, or environmental compliance may increase production costs and alter market dynamics.
Bike-Sharing Expansion: The rise of bike-sharing programs in cities offers opportunities for manufacturers and service providers to meet the demand for shared mobility solutions.	**Substitution Risks**: Competing transport alternatives, such as electric scooters, public transit, or ride-sharing platforms, may reduce demand for bicycles.
Eco-Tourism Potential: The increasing popularity of sustainable tourism creates opportunities for bicycle rental services and tour operators in tourist destinations.	**Supply Chain Challenges**: Global supply chain disruptions, including shortages of raw materials or logistical issues, can hinder bicycle production and distribution.
Customized Offerings: Growing consumer interest in personalized and bespoke bicycles	**Competitive Pressures**: Intense competition from both established manufacturers and new entrants may

allows manufacturers to cater to individual preferences, enhancing market appeal.	lead to price wars, margin erosion, and market saturation.

1.1.2.2. Business Problems

Through an evaluation of the global bicycle industry and the application of the Business Model Canvas, several key challenges facing AdventureWorks Cycles (AWC) have been identified:

- **Competitive Landscape and Brand Positioning**: The bicycle industry's intense competition presents difficulties for AWC in differentiating its products and establishing a unique brand identity amidst numerous competitors.
- **Aligning with Customer Needs and Trends**: To effectively engage and retain its customer base, AWC must adapt to evolving consumer preferences, particularly those driven by health, fitness, and urban lifestyle trends. This necessitates robust customer segmentation and tailored analysis, which are critical components of the Business Model Canvas.
- **Expansion into New Markets**: Entering emerging markets in regions such as Asia, Latin America, and Africa offers growth potential for AWC. However, success in these markets requires comprehensive market research, strategic planning, and adaptation to local regulations and consumer behaviors.
- **Seasonal Sales Variability**: AWC faces sales fluctuations due to seasonal factors, such as lower demand during colder or wetter periods, requiring effective strategies for inventory and production management. This issue aligns with the Business Model Canvas emphasis on demand forecasting to mitigate risks of overstock or shortages.
- **Market Saturation and Price Competition**: The highly competitive nature of the industry may result in market saturation, exerting downward pressure on pricing and potentially affecting AWC's profitability and market share.
- **Technological Innovation and Adaptation**: To remain competitive, AWC must stay at the forefront of advancements in bicycle design, materials, and technology to meet changing customer expectations and preferences.
- **Supply Chain Efficiency**: Disruptions in the global supply chain, such as raw material shortages or logistical challenges, could hinder AWC's production and distribution capabilities, impacting its ability to meet customer demand and delivery timelines.

Focusing specifically on the Purchasing domain, the research team identified four primary challenges with

1.2 Objectives of the project

1.2.1 General Objective

The primary aim of deploying Business Intelligence (BI) solutions at AdventureWorks is to empower data-driven decision-making and foster a culture centered on data utilization through the adoption of advanced analytical tools. BI serves as a strategic instrument, delivering a holistic perspective on both historical and real-time data to provide actionable insights across various business operations. The overarching goal is to enhance operational effectiveness and enable agile responses to evolving market dynamics. By leveraging BI tools, such as interactive dashboards and detailed reports, the initiative seeks to streamline procurement processes, reduce inefficiencies and costs, and improve organizational performance. Additionally, integrating data across departments promotes cross-functional collaboration, enhancing coordination among purchasing, sales, inventory management, and production teams, thereby cultivating a unified and efficient work environment.

1.2.2 Specific Objectives

- **Business Requirements Analysis**: Examine the operational needs and challenges across departments, pinpoint key performance indicators (KPIs) and metrics related to inventory and purchasing, and evaluate existing processes to identify inefficiencies and opportunities for enhancement.
- **Data Source Evaluation**: Perform a thorough review of available data sources within AdventureWorks, assessing their quality, completeness, and relevance while identifying any inconsistencies or gaps that require resolution.
- **Data Warehouse and Data Mart Development**: Design a robust data warehouse architecture and specialized data marts to efficiently store and structure data for analytical and reporting purposes.
- **Data Integration into Data Model and Warehouse**: Execute seamless data integration processes to consolidate information from diverse sources into a unified data model and warehouse infrastructure.
- **Data Visualization Implementation**: Employ visualization tools, such as dashboards and reports, to present key metrics and insights derived from BI analysis in an accessible and impactful manner, supporting informed decision-making.
- **BI Solution Recommendations**: Propose customized BI solutions tailored to AdventureWorks' specific requirements, ensuring scalability, adaptability

1.3 Research Objects

The primary aim of this study is to analyze the operational dynamics of the purchasing department within Adventure Works Cycles, a fictional multinational bicycle

manufacturer, to design a business intelligence (BI) solution that enhances strategic decision-making. By leveraging data-driven insights, the project seeks to optimize procurement processes and improve organizational efficiency. The research encompasses the following key components:

- **Company Context**: The BI solution is tailored for Adventure Works Cycles, a fictional entity utilized for analytical purposes.
- **Data Source**: The analysis draws upon the AdventureWorks 2019 database, providing a comprehensive dataset for procurement-related insights.
- **Database Platform**: SQL Server serves as the foundational technology for data storage and querying.
- **Business Process**: The study focuses on the procurement process, examining its workflows and operational intricacies.
- **Development Environment**: Microsoft Visual Studio is employed for building and integrating the BI solution.
- **Analytical Technique**: Online Analytical Processing (OLAP) is utilized for multidimensional data analysis to uncover actionable insights.
- **Performance Metrics**: Key Performance Indicators (KPIs) are developed to evaluate procurement efficiency and effectiveness.
- **Visualization Tool**: Power BI is implemented to create intuitive, interactive visualizations for enhanced decision-making support.

By integrating these components, the research aims to deliver a robust BI framework that empowers Adventure Works Cycles to make informed, data-driven decisions, thereby strengthening its procurement operations and competitive position.

1.4. Scope of the Project

- **Time Scope**: The research will be conducted over the period from July 15, 2025, to September 30, 2025, allowing for comprehensive data collection and analysis within this timeframe.
- **Dataset Time Scope**: The analysis will leverage historical data from the AdventureWorks 2019 database to establish a robust foundation for identifying trends and patterns in procurement activities.
- **Spatial Scope**: The study centers on Adventure Works Cycles, with a particular emphasis on the operations of its Purchasing Department, examining processes, workflows, and performance metrics to inform the BI solution.

1.5. Value and desired outcome of the project

The initiative by AdventureWorks to develop a data warehouse for its sample database is anticipated to deliver substantial benefits and align with the company's strategic

objectives. This data warehouse will enhance AdventureWorks' analytical and reporting capabilities, enabling the generation of precise, timely, and actionable insights. Such improvements will facilitate informed decision-making, positioning the company to maintain a competitive edge in the marketplace. Furthermore, the implementation of a data warehouse will minimize errors and inconsistencies in data reporting, thereby increasing the reliability and accuracy of critical business information. Consequently, AdventureWorks will be better equipped to leverage data-driven strategies, ultimately enhancing operational performance and market competitiveness.

The project is also expected to yield efficiencies in both time and cost. By optimizing data processing and analytical workflows, the data warehouse will reduce the resources and effort required for preparing data, thereby improving overall operational efficiency and productivity.

In summary, the data warehouse initiative is poised to provide AdventureWorks with a robust and efficient platform to meet its reporting and analytical requirements. By delivering enhanced data integration, improved accessibility, reduced operational costs, and increased efficiency, the project will empower the company to make well-informed decisions and achieve its strategic goals, thereby creating significant value for the organization.

1.6 Tools, Methods, and Programming Languages

1.6.1 SQL Server

Microsoft SQL Server is a relational database management system (RDBMS). Applications and tools connect to a SQL Server instance or database, and communicate using Transact-SQL (T-SQL).

1.6.2 Microsoft Visual Studio

The Visual Studio IDE is a creative launching pad that you can use to edit, debug, and build code, and then publish an app. Over and above the standard editor and debugger that most IDEs provide, Visual Studio includes compilers, code completion tools, graphical designers, and many more features to enhance the software development process.

1.6.3 Multidimensional Data Analysis Technique (OLAP)

Online analytics processing (OLAP) is the software technology you can use to analyze business data from different perspectives. Organizations collect and store data from a

variety of data sources, such as websites, applications, smart watches and internal systems. OLAP combines and groups this data into categories to provide actionable details for strategic planning. For example, a retailer stores data about all the products it sells, such as color, size, cost and location. The retailer also collects customer purchase data, such as the names of ordered items and the total sales value, in another system. OLAP combines data sets to answer questions like which color products are more common or how product placement affects sales.

OLAP patterns are a means of documentation for best-practice solutions regarding query composition—with support for automatic query generation. Therefore, the proposed pattern-based approach to multidimensional data analysis distinguishes two separate activities: Pattern definition and pattern usage. Pattern definition comprises bottom-up specification of patterns in the context of a specific data warehouse system as well as top-down derivation of more specific patterns from more general patterns in a catalog. Pattern usage, on the other hand, requires the pattern user to bind values to pattern parameters, which allows for the subsequent grounding and execution of the pattern in the context of a specific data warehouse system. Different business intelligence users are involved in the course of pattern definition and usage. While pattern definition will typically involve pattern authors with a profound BI understanding at least to some extent, such as IT developers, data engineers, and data scientists, pattern usage directly concerns the pattern users with a basic BI understanding, such as decision makers, domain experts, business analysts, and power users. In this sense, OLAP patterns serve to communicate knowledge of best-practice solutions for satisfying generic types of information needs from experienced pattern authors to pattern users. It should be noted that a person can assume both the role of a pattern author and pattern user.

1.6.4. Power BI

Power BI is a data analytics and visualization platform developed by Microsoft, playing a pivotal role in the modern Business Intelligence (BI) ecosystem. With its ability to connect to a wide range of data sources—such as SQL Server, Excel spreadsheets, cloud-based services, and enterprise management systems—Power BI enables users to integrate data from multiple origins and generate a comprehensive, visual overview of organizational performance.

Its user-friendly and intuitive interface allows users, including those without advanced technical backgrounds, to easily design analytical reports and dashboards through a drag-and-drop approach. Power BI also incorporates powerful data processing tools, such as Power Query for data cleaning and transformation, as well as DAX (Data Analysis Expressions), a functional language that facilitates complex calculations and in-depth data modeling. As a result, charts, maps, KPIs, and summaries built in Power BI are not only visually engaging but also analytically meaningful.

Beyond data presentation, Power BI excels in providing high interactivity across visual components. Users can dynamically filter data, drill down into details, and observe trends within a single, unified report. Furthermore, seamless integration with platforms such as Microsoft Teams, SharePoint, and Power BI Service enhances the accessibility and collaborative use of insights across organizational departments.

Within the scope of this project, Power BI was employed as the primary visualization tool to present analytical results derived from the data warehouse and OLAP models. Through Power BI, the project team was able to communicate key insights effectively, thereby supporting instructors, decision-makers, and stakeholders in better understanding emerging trends, challenges, and opportunities embedded in the data.

1.7. Structure of project

The project is organized into six chapters, each addressing a critical aspect of the development and implementation process:

- **Chapter 1: Introduction**
  This chapter presents an overview of the project, describes the AdventureWorks sample database, and articulates the project's objectives. It also delineates the scope and provides essential contextual details.
- **Chapter 2: Theoretical Basis**
  This chapter examines the foundational theories and concepts underpinning the Business Intelligence (BI) solution. It encompasses key topics such as data warehousing, data integration, and data analytics, establishing a comprehensive theoretical framework.
- **Chapter 3: Requirements Analytics and Introduction to BI Solution**
  This chapter assesses the project's requirements and introduces the proposed Business Intelligence solution, emphasizing its alignment with identified needs and goals.
- **Chapter 4: Building Data Warehouse and Integrating Data**
  This chapter elaborates on the development of the data warehouse and the integration of diverse data sources. It provides a detailed account of the ETL

(Extract, Transform, Load) process and the tools and technologies utilized in constructing the data warehouse.

- **Chapter 5: Results - Data Analytics and Visualization**
  This chapter showcases the project's outcomes, focusing on data analysis and visualization achieved through tools such as Power BI. It evaluates the insights derived and their potential to support strategic decision-making for AdventureWorks.
- **Chapter 6: Conclusion and Future Works**
  This chapter summarizes the project's findings and reflects on the implementation process. It also offers recommendations for future enhancements and potential expansions of the project.

1.8. Research Process

The research process for the project is structured into nine distinct steps, as outlined below:

- **Requirement Identification**
  The team conducts an in-depth analysis to understand the objectives of enhancing the purchasing function within the AdventureWorks database. This involves reviewing relevant literature, assessing evaluation criteria and methodologies, and defining the content requirements to establish a clear direction for the Business Intelligence (BI) solution.
- **Database Selection and Analysis**
  This step focuses on identifying and procuring a suitable database that aligns with the project's objectives. The team thoroughly examines the AdventureWorks database, analyzing its overall structure and the specific data related to the purchasing function to ensure its relevance and applicability.
- **Project Overview Development**
  A comprehensive overview of the project is formulated, encompassing industry context, purchasing fundamentals, business case, objectives, research scope, budget, deliverables, processes, tools, methodologies, programming languages, and project structure. This ensures alignment and clarity among all stakeholders.
- **Exploration of Relevant Theories and Concepts**
  The team investigates key theoretical frameworks and concepts related to Business Intelligence, including ETL (Extract, Transform, Load) processes, data warehousing, data marts, and key performance indicators (KPIs). This theoretical foundation equips the team to address project requirements effectively.

- **Requirements Analysis and BI Solution Proposal**
  This phase involves a detailed evaluation of business processes, requirements, and KPIs to determine both functional and technical needs. Based on this analysis, a customized BI solution is proposed to address the challenges associated with the purchasing function in the AdventureWorks database.
- **Data Source Identification, Data Warehouse Design, and Integration**
  The team identifies appropriate data sources, designs an efficient data warehouse architecture, and develops data marts tailored to specific analytical requirements. Data from various sources is integrated seamlessly to support a cohesive BI solution.
- **Data Analysis and Visualization**
  Reporting systems and dashboards are developed, leveraging tools such as Power BI to analyze and visualize data. This step provides stakeholders with clear, actionable insights to enhance the purchasing function within the AdventureWorks database.
- **Conclusions and Future Directions**
  The project's outcomes are summarized, highlighting key findings and limitations. Recommendations for future improvements or expansions are proposed to ensure the project's long-term impact and relevance.
- **Project Management Evaluation**
  A thorough review of the project management process is conducted, assessing performance against established metrics, evaluating the overall process, and analyzing costs to identify opportunities for optimization in future initiatives.

Through a structured and methodical approach involving rigorous research, strategic analysis, and the implementation of a tailored BI solution, the team effectively addressed the challenges of the purchasing function within the AdventureWorks database. This effort has established a foundation for improved efficiency and informed decision-making while identifying avenues for further refinement and development.

CHAPTER 2: THEORETICAL BASIS

2.1. Overview about BI

2.1.1. What is BI?

Business Intelligence (BI) refers to a technology-driven process for analyzing data and delivering actionable information to support informed decision-making across all levels of an organization. It integrates various components, including business analytics, data mining, data visualization, data infrastructure, and established best practices, with the overarching aim of fostering data-informed organizational strategies.

The BI process typically begins with the collection and preparation of relevant data, followed by the execution of analytical queries. The resulting insights are employed to guide both operational and strategic decision-making processes. Through this systematic approach, BI seeks to enhance organizational performance and responsiveness.

Historically, the concept of Business Intelligence has undergone significant evolution. Its roots trace back to the 1960s, when early systems were developed to facilitate the dissemination of information across enterprises. The term "Business Intelligence" was formally introduced in 1989, coinciding with the emergence of computer-based decision support models. Over time, these models evolved into specialized BI solutions characterized by IT-centric service delivery and the transformation of data into strategic insights.

The principal objective of BI initiatives is to improve business outcomes—namely, by increasing revenue, optimizing operational efficiency, and gaining competitive advantage. This is accomplished through the effective application of analytical methodologies, data visualization techniques, and comprehensive reporting tools.

2.1.2. BI Architecture

Business Intelligence (BI) Architecture is a comprehensive framework that underpins the effective collection, storage, analysis, and dissemination of data within an organization. Serving as a structured blueprint, BI architecture ensures that relevant data is delivered to the appropriate stakeholders in a timely manner, thereby enabling informed, data-driven decision-making across all levels of the enterprise.

At its foundation, BI architecture involves the strategic design and deployment of data infrastructures, processes, and technologies that support business intelligence activities. It comprises multiple interconnected components, including data sources, data integration, data storage, data modeling, data analysis, and data visualization. These components are built upon four core conceptual pillars: Data Mining and Analysis, Reporting and Visualization, Performance Management and Benchmarking, and Predictive Analytics.

Core Components of BI Architecture

1. Data Sources:

The BI process begins with the identification and integration of data from diverse internal and external sources, such as transactional databases, spreadsheets, enterprise applications, and cloud-based platforms. The goal is to create a unified and comprehensive data landscape that accurately reflects the organization's operations.

2. Data Integration:

Once data sources are identified, the next step involves transforming and consolidating data into a standardized format. Data integration encompasses processes such as cleansing, enrichment, and transformation to ensure consistency, accuracy, and quality throughout the organization's data ecosystem.

3. Data Storage:

BI architecture relies on robust data storage solutions that may range from traditional relational databases to modern big data platforms. These systems must support high data volumes and offer performance, scalability, and reliability to accommodate evolving analytical needs.

4. Data Modeling:

Data modeling defines the logical structure and relationships among data elements, serving as the foundation for efficient data retrieval and analysis. Common data modeling techniques include the use of star and snowflake schemas, which organize data into intuitive, analytical formats.

5. Data Analysis:

Analytical tools and techniques are employed to extract insights and uncover patterns from data. These may include basic aggregations and descriptive statistics as well as more advanced methodologies, such as machine learning algorithms and statistical modeling, to support deeper analysis.

6. Data Visualization:

The final stage involves presenting analytical results in a clear and accessible manner. Visualization tools transform complex data into intuitive dashboards, graphs, and charts that facilitate quick understanding, pattern recognition, and informed decision-making.

Core Concepts of BI Architecture

Data Mining and Analysis:

Data mining involves examining large datasets to discover hidden patterns, correlations, and trends. Organizations use data mining to better understand customer behaviors, optimize marketing strategies, enhance sales performance, and reduce operational costs.

Reporting and Visualization:

Reporting transforms raw data into structured outputs such as dashboards and performance summaries, while visualization employs graphical representations—such as charts, maps, and infographics—to present data in a user-friendly format. Together, they enhance the interpretability and communicability of business insights.

Performance Management and Benchmarking:

Performance management refers to the continuous process of setting organizational goals, monitoring progress, and using feedback to improve outcomes. Benchmarking complements this by comparing performance metrics against industry standards or peer organizations to identify best practices and areas for improvement.

Predictive Analytics:

Predictive analytics applies statistical techniques and machine learning to historical data in order to forecast future trends and behaviors. This forward-looking approach allows organizations to anticipate challenges, capitalize on emerging opportunities, and make proactive strategic decisions.

By implementing a well-structured Business Intelligence Architecture, organizations can unlock the full value of their data assets. This enables improved strategic planning, operational efficiency, market responsiveness, and sustainable business growth—ultimately offering a significant competitive advantage in data-driven environments.

2.1.3. Advantage of BI in enterprises

There are many reasons why organizations adopt business intelligence and analytics tools, and many ways these solutions are put to work to benefit the organization. All projects, however, have the common goal to use business intelligence software to turn data into insights and action.

As the primary competitive advantage business intelligence should deliver an increase in the understanding of the forces shaping markets and businesses, and help companies to act on that knowledge.

A list of some of the potential benefits of business intelligence and were asked to indicate to what level they had achieved those benefits with BI:

Faster reporting, analysis or planning, More accurate reporting, analysis or planning

Trend insights

1. Enhanced accuracy and reliability of data
2. Increased employee engagement and job satisfaction
3. Optimized operational performance and resource utilization
4. Elevated customer experience and satisfaction levels
5. Strengthened market positioning and competitive differentiation
6. Lowered operational and overhead expenses
7. Expanded revenue streams and business profitability
8. Reduced staffing requirements through process automation
9. Facilitated data-driven decision-making leading to improved strategic outcomes

The following chart shows where BI software really helps companies.

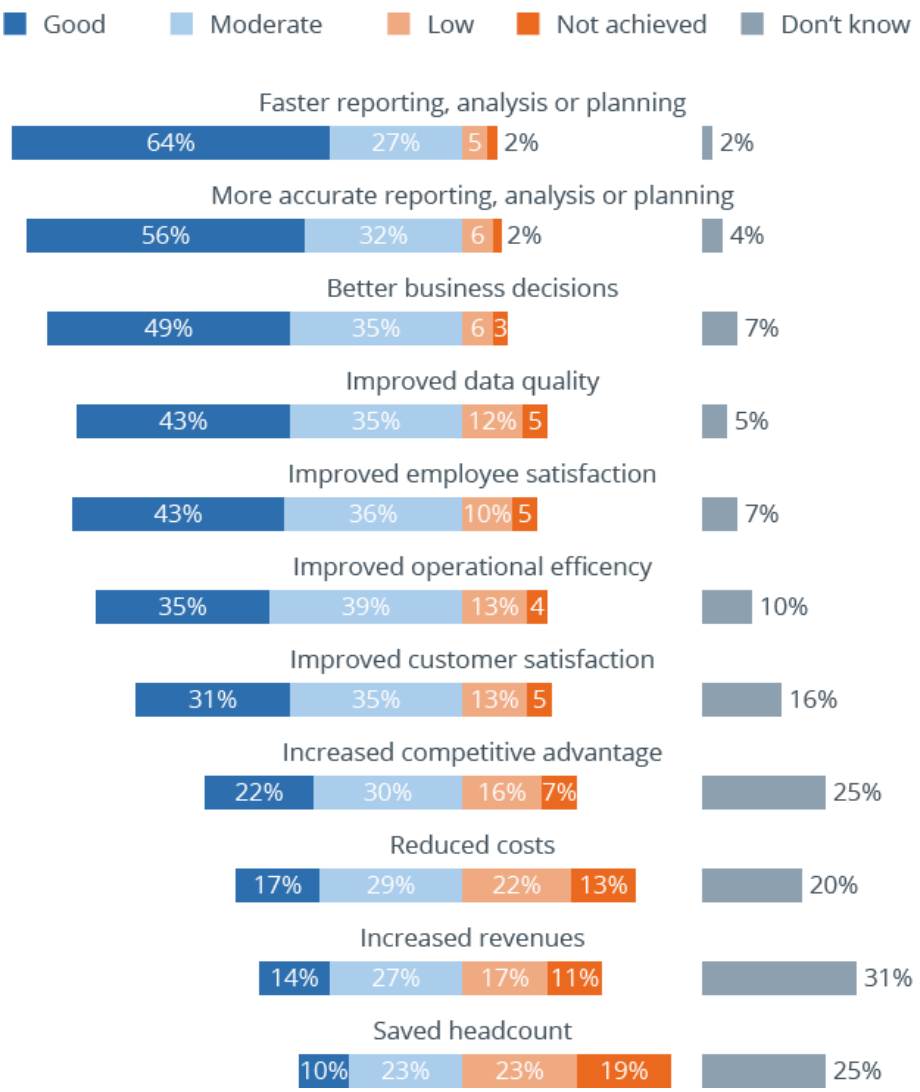


Figure 2.1. To what level have you achieved the following benefits? (Sourrce: BARC)

The chart illustrates the most commonly cited benefits of Business Intelligence (BI) tools, ranked by frequency. Foremost among these, BI and IT leaders view their platforms as essential for delivering faster and more accurate information to decision-makers. The most frequently achieved outcomes include accelerated and more precise reporting, analysis, and planning; enhanced decision-making; increased employee satisfaction; and improved data quality.

Conversely, the least commonly realized benefits are cost reduction, revenue growth, and reductions in IT staffing. However, as organizations increasingly seek to monetize their data through enhanced product and service offerings, it is anticipated that the frequency of reported revenue gains from BI initiatives will rise in the future

2.2. ETL Process

2.2.1. What is ETL?

ETL, which stands for Extract, Transform, and Load, is a data integration process that involves retrieving data from various sources, cleansing and transforming it into a consistent format, and then loading it into a centralized storage system such as a data warehouse or data lake. This process ensures that data from disparate systems is harmonized and ready for analysis.

An ETL data pipeline serves as the foundational layer for data analytics and machine learning workflows. It performs three essential functions:

- Data Extraction – It retrieves data from legacy or source systems.
- Data Transformation – It enhances data quality through cleaning, standardization, and formatting to ensure consistency and reliability.
- Data Loading – It stores the refined data into a target system or modern database, making it accessible for analytical and business intelligence purposes.

Ultimately, ETL processes ensure that data is accurate, consistent, and readily available for meeting business objectives, thereby supporting more efficient and high-quality decision-making.

2.2.2. Why do we need ETL?

ETL is crucial for businesses looking to consolidate large volumes of data, thereby improving decision-making and providing reliable insights. In today's digital landscape, where data is sourced from diverse origins and formats, the ETL process ensures that organizations can effectively clean, standardize, and organize this information for advanced analytics. It creates a structured foundation for data analysis, enhancing the quality, security, and accessibility of enterprise data.

2.2.3. ETL Process

The ETL process, an acronym for Extract, Transform, and Load, represents a fundamental methodology employed in the preparation of data for storage, analysis, and reporting within a data warehouse. This process is comprised of three distinct stages, each designed to facilitate the systematic transformation of raw data from various sources into a clean, structured, and usable format. Below is a detailed examination of each phase:

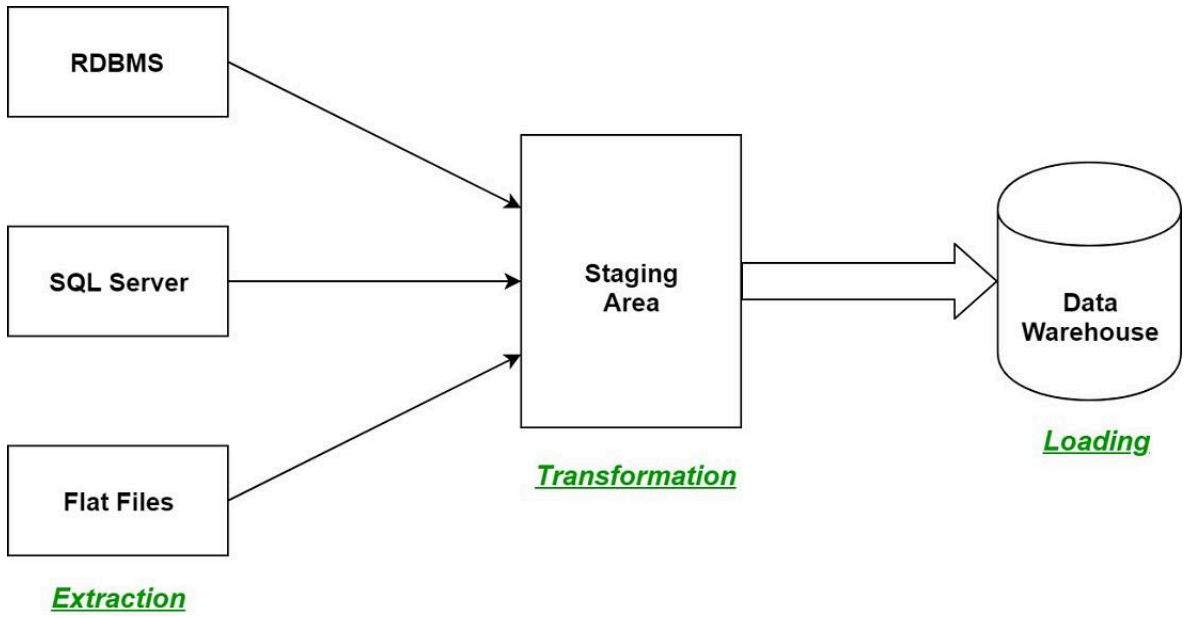


Figure 2.2. The ETL simple process (Source by Geek4Geek)

2.2.3.1. Extraction

The Extract phase constitutes the initial step in the ETL process, during which raw data is gathered from a variety of data sources. These sources can be categorized into several types, including structured sources such as SQL and NoSQL databases, semi-structured formats like JSON and XML, and unstructured data, which encompasses emails and flat files. The primary objective of the extraction process is to collect data without modifying its original format, thereby facilitating its subsequent processing in the next phase.

Types of data sources include:

- Structured: SQL databases, Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) systems
- Semi-structured: JSON, XML
- Unstructured: Emails, web pages, flat files

2.2.3.2. Transformation

The Transform phase is a critical stage in the ETL process—where raw, inconsistent data is refined and structured to meet analytical and business requirements. After extraction, data is often unclean and lacks uniformity. During transformation, it undergoes cleaning, aggregation, and reformatting based on predefined business rules.

This step is essential to ensure data quality, consistency, and reliability for accurate downstream analysis.

Typical transformation operations include:

- **Data Filtering**: Eliminating irrelevant, duplicate, or erroneous records.
- **Data Sorting**: Arranging data in a specified order to facilitate more efficient analysis.
- **Data Aggregation**: Summarizing data, such as calculating averages or totals, to generate higher-level insights.

Additionally, the transformation phase may involve more advanced tasks such as currency conversion, standardizing text formats, or applying domain-specific logic to align the data with organizational standards and analytical goals.

2.2.3.3. Loading

Upon completion of data cleaning and transformation, the final step in the ETL process is Loading. This phase entails transferring the transformed data into a designated storage system, such as a data warehouse or data lake. Depending on the specific use case, there are two primary loading methods:

Full Load: This method involves loading all data into the target system, typically employed during the initial population of the data warehouse.

Incremental Load: In contrast, this method only loads new or updated data, thereby enhancing efficiency for ongoing data updates.

2.2.3.4. Importances of ETL process

In today's data-driven environment, organizations generate and depend on vast amounts of data to guide strategic and operational decisions. ETL (Extract, Transform, Load) offers a streamlined approach to managing, accessing, and utilizing this data effectively. When implemented properly, ETL brings several key advantages:

Historical Perspective: ETL enables organizations to retain both current and legacy data within centralized repositories. This comprehensive view allows for comparisons across different time periods, helping companies trace their evolution, understand long-term trends, and make informed decisions in areas such as marketing, production, and customer engagement.

Comprehensive Data Integration: ETL facilitates the consolidation of data from multiple sources—ranging from structured databases to unstructured formats like logs or documents—into a unified and coherent system, supporting holistic analysis.

Improved Data Quality: Through the transformation phase, ETL processes clean, standardize, and validate raw data. This ensures higher levels of accuracy, consistency, and reliability, forming a strong foundation for trustworthy analytics.

Support for Data Warehousing: ETL is a foundational element in preparing data for storage in data warehouses. It ensures that data conforms to the schema and structure required by the target system, enabling efficient querying and reporting.

Better Decision-Making: By organizing and enhancing data for analytical use, ETL enables businesses to extract meaningful insights. These insights support more accurate forecasting, better resource allocation, and more effective strategic planning.

Increased Operational Efficiency: Automating data processing through ETL pipelines accelerates data availability and reduces manual effort. This enables organizations to respond more quickly to changing conditions and make timely, data-informed decisions.

2.3. Data warehouse and Data mart

2.3.1. What are Data warehouse and Data mart?

2.3.1.1. Data Warehouse

A data warehouse is a specialized data management system designed to support business intelligence (BI) activities, with a particular focus on analytical tasks. Unlike transactional systems, data warehouses are optimized for querying and analysis, and they often contain vast volumes of historical data sourced from various origins—such as transactional applications, application log files, and external systems.

At its core, a data warehouse centralizes and consolidates data from multiple heterogeneous sources, providing a unified platform for analysis. Its robust analytical capabilities enable organizations to extract meaningful business insights, enhancing decision-making processes. Over time, it accumulates historical data that is particularly valuable for business analysts and data scientists. Due to its role in

maintaining consistent and reliable information, a data warehouse is often regarded as the organization's "single source of truth."

A typical data warehouse architecture includes several key components:

- A relational database for storing and managing structured data.
- An Extract, Load, and Transform (ELT) or ETL pipeline to clean, prepare, and load data into the warehouse for analysis.
- Tools for statistical analysis, reporting, and data mining to uncover trends and patterns.
- Client-facing analytical tools for data visualization, dashboards, and business reporting.
- Advanced analytics capabilities, which may include artificial intelligence (AI), machine learning (ML), graph analytics, or spatial analysis for deeper, large-scale data exploration.

Modern solutions are increasingly integrating transactional processing, real-time analytics, and machine learning into unified platforms—eliminating the need for separate systems and reducing the complexity, latency, cost, and risk associated with traditional ETL duplication. For example, some MySQL-based services now support seamless integration across data warehouses, data lakes, and advanced analytics tools, offering a holistic and agile data ecosystem.

2.3.1.2. Data mart

A data mart is a focused subset of a data warehouse, tailored to serve the needs of a specific business function, department, or subject area. By narrowing the scope of data to a particular domain—such as finance, sales, or marketing—data marts enhance team productivity, reduce operational costs, and support more effective and timely tactical decision-making.

Data marts provide targeted access to relevant datasets for defined user groups, enabling them to quickly retrieve critical insights without navigating the broader enterprise data warehouse. This streamlined access minimizes delays and enhances the efficiency of analytical processes.

Although data marts may still house millions of records, they are purpose-built to deliver highly relevant information within a limited context. Their primary goal is to

empower business users with fast, reliable access to the data most pertinent to their roles—supporting focused analysis and decision-making.

2.3.2 Snowflake and Star schemas?

A data mart is a subject-oriented relational database that stores transactional data in rows and columns, which makes it easy to access, organize, and understand. As it contains historical data, this structure makes it easier for an analyst to determine data trends. Typical data fields include numerical order, time value, and references to one or more objects.

Companies organize data marts in a multidimensional schema as a blueprint to address the needs of the people using the databases for analytical tasks. The three main types of schema are star, snowflake, and galaxy.

2.3.2.1. Star schema

Star schema is a logical formation of tables in a multidimensional database that resembles a star shape. In this blueprint, one fact table—a metric set that relates to a specific business event or process—resides at the center of the star, surrounded by several associated dimension tables.

There is no dependency between dimension tables, so a star schema requires fewer joins when writing queries. This structure makes querying easier, so star schemas are highly efficient for analysts who want to access and navigate large data sets.

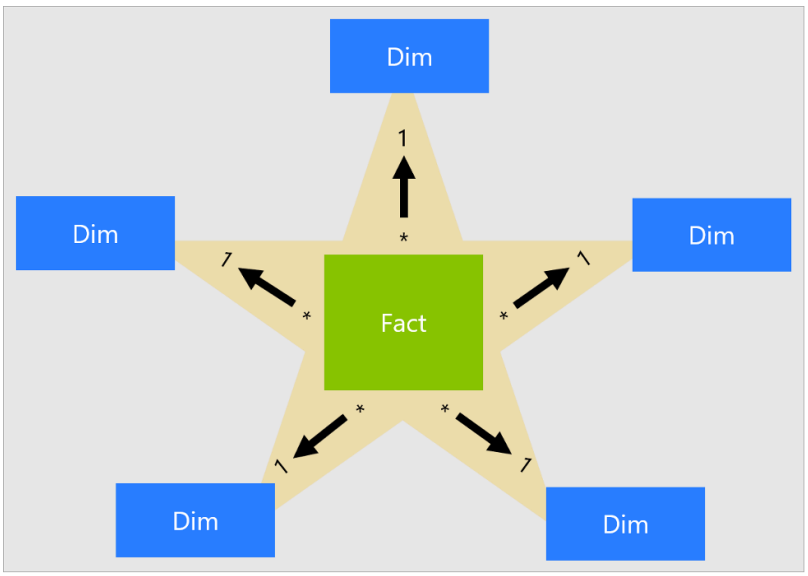


Figure 2.3. Star schema (source by Authors)

2.3.2.2. Snowflake schema

A snowflake schema is a multi-dimensional data model that is an extension of a star schema, where dimension tables are broken down into subdimensions. Snowflake schemas are commonly used for business intelligence and reporting in OLAP data warehouses, data marts, and relational databases.

In a snowflake schema, engineers break down individual dimension tables into logical subdimensions. This makes the data model more complex, but it can be easier for analysts to work with, especially for certain data types.

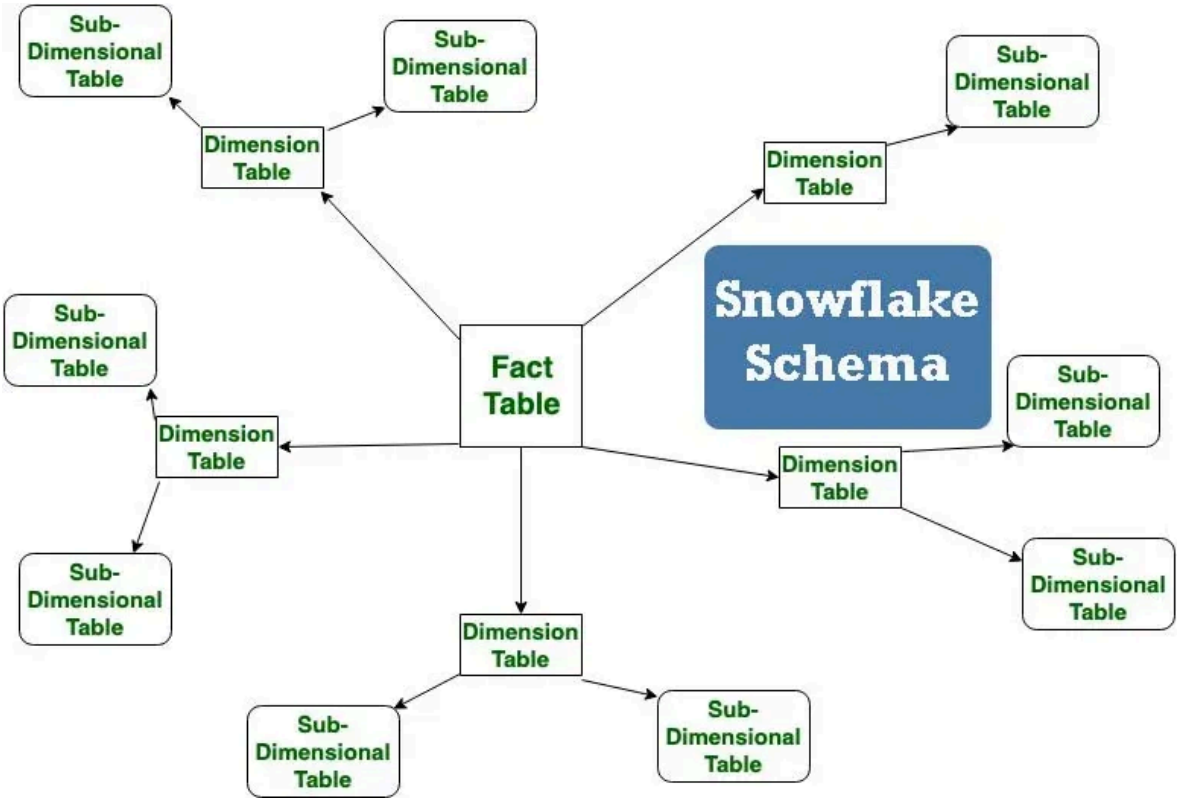


Figure 2.4. Snowflake schema (Source by Geeks4Geeks)

2.3.2.3. Galaxy Schema

The galaxy schema is a hybrid design that combines elements of both the star and snowflake schemas. It allows for more flexibility and complexity by connecting multiple fact tables to multiple dimension tables. This can be useful when dealing with multiple business processes or areas.

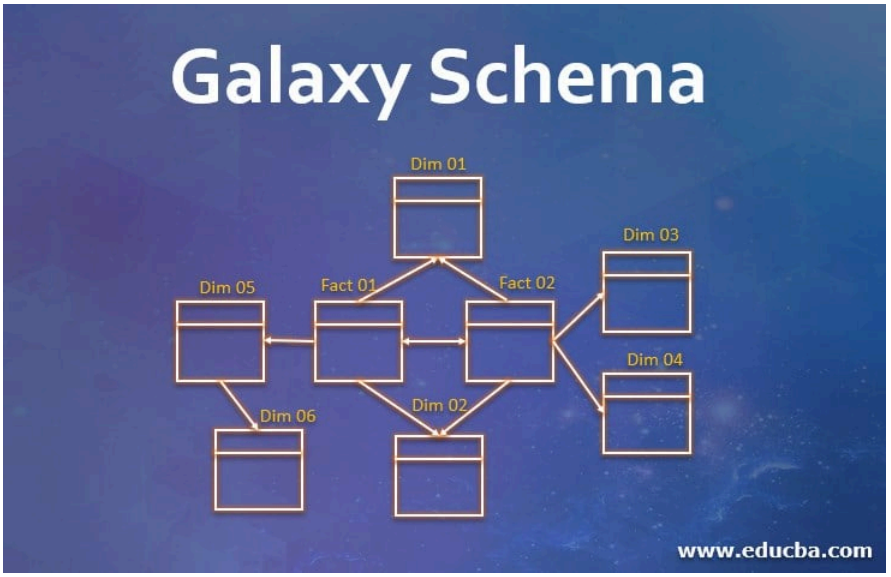


Figure 2.5. Galaxy schema (Source: educuba)

2.3.3. Who needs Data warehouse and Data mart?

Every organization, regardless of its industry or size, needs a data warehouse to integrate its various data sources. This integration supports anticipation, analysis, reporting, business intelligence, and informed decision-making. Additionally, these services should be delivered at a reasonable cost while providing optimal value.

2.3.4. Advantages and disadvantages of Data warehouse

2.3.4.1. Advantages

Throughout Chapter 2, we have explored various benefits of data warehousing. In addition to the advantages already discussed, several key benefits further highlight the strategic value that data warehouses offer to organizations:

Enhanced Business Intelligence

One of the primary advantages of a data warehouse is its ability to significantly improve business intelligence. By providing a centralized, unified view of data sourced from multiple systems, data warehouses simplify access to critical information. This integration enables more efficient and effective analysis, supporting data-driven decision-making across the organization. As a result, complex data management tasks are streamlined, and insights become more readily accessible.

Improved Data Quality and Consistency

Maintaining high standards of data quality and consistency is essential for reliable analytics. Data warehouses enforce rigorous data cleansing and transformation processes that eliminate inconsistencies, duplicates, and inaccuracies. This ensures that the data stored within the system is both accurate and reliable, allowing stakeholders to trust the insights generated from it.

Cost and Time Efficiency

Data warehouses contribute to organizational efficiency by automating and optimizing the processing and analysis of large datasets. This reduces the need for manual data handling, saving both time and operational costs. Employees can redirect their efforts from routine data preparation to more strategic, value-added activities, thereby improving overall productivity.

Access to Historical Data for Strategic Analysis

Data warehouses maintain extensive historical records, which are invaluable for identifying trends, evaluating performance over time, and forecasting future outcomes. This historical perspective supports long-term strategic planning and helps organizations respond proactively to emerging opportunities and challenges.

High Return on Investment (ROI)

Investing in a data warehouse can yield substantial returns. By ensuring data quality, enabling comprehensive analytics, and streamlining data integration, organizations gain actionable insights that lead to better decisions, reduced costs, increased revenue, and improved operational performance. The competitive advantage gained through these capabilities translates directly into measurable business value.

2.3.4.2. Disadvantages

However, there will have some disadvantages, such as:

Additional Reporting

Additional work is required for using data warehouses because data stored in a warehouse is structured, meaning that to create reports, we must design and maintain predefined queries and data models. This is time-consuming and delays in generating essential reports.

Inflexibility and Homogenization of Data

Data warehouses depend on structured data, which is organized into predefined formats. This can be a problem when handling unstructured data, such as social media posts or multimedia content. Unstructured data doesn't fit well within the structured framework of a data warehouse. If the organization deals with a lot of unstructured data, it will face problems in integrating the data effectively.

Ownership Concerns

Data warehousing systems often involve multiple departments and teams. This can lead to ownership issues. Determining who is responsible for various aspects of the data warehouse, including data governance and quality, can become a complex task. Lack of clear ownership can result in inefficiencies and data management problems.

Demands for Large Amounts of Resources

Implementing and maintaining a data warehouse can be resource-intensive. The organization has to invest in hardware, software, and skilled personnel to set up and manage the system. This upfront investment can be substantial and may not be feasible for small or resource-constrained organizations. The ongoing costs of maintaining a data warehouse can also add up over time.

Hidden Issues Consume Time

Data warehouses are not immune to hidden problems. Issues related to data quality, transformation errors, and data integration challenges can emerge, and uncovering these problems can consume valuable time and resources. Such hidden issues can also lead to inaccurate reports and analyses, affecting decision-making.

2.4. KPIs

2.4.1. KPIs Definition

Key Performance Indicators (KPIs) are measurable metrics used to assess an organization's overall performance over time. They serve as vital tools for tracking progress toward strategic business objectives and evaluating success across various domains, including financial, operational, and organizational goals.

KPIs provide insight into how well a company is performing in relation to its targets, industry standards, or historical performance. By comparing these indicators internally over time or externally against competitors, organizations can identify strengths,

address weaknesses, and make data-informed decisions to drive continuous improvement.

2.4.2. Categories of KPIs

There are a variety of KPIs that can help measure the performance or progress of your business. Some types of KPIs include:

1. **Quantitative indicators**: Quantitative indicators are expressed as continuous or discrete numbers, including ratios, percentages, or whole numbers that reflect values such as rating scales, monetary amounts, or weight. These indicators serve as the most direct and measurable metrics of performance, providing clear numerical values for analysis.
2. **Qualitative indicators**: These indicators are conveyed not through numerical values but rather through feelings or opinions. For instance, an employee satisfaction survey exemplifies qualitative data, where performance is assessed based on respondents' feedback.
3. **Leading indicators**: Leading indicators are metrics that can reveal long-term trends and may predict positive future outcomes for business processes.
4. **Lagging indicators**:Lagging KPIs assess a business's current performance in a specific area by comparing it to its past performance in that same area.
5. **Input indicators**: Input indicators are a type of KPI that track the resources necessary to produce the intended outcome, such as funding or extra staff. Input indicators can help companies keep track of how efficiently they are using their resources.
6. **Output indicators**: Output indicators measure the success or failure of your business activities, like the number of goods or services created through a particular process. Revenue growth and new customer acquisition also indicate how well your business is performing.
7. **Process indicators**: Process indicators represent the efficiency of a business’s process and how effectively it is functioning.
8. **Practical indicators**: Practical indicators explore the function of an existing process at a company, usually involving observation or feedback on that process.

9. **Directional indicators**: Directional indicators help determine the company's success in comparison with competitors, while practical indicators are specific to the company's process within itself.
10. **Actionable indicators**: Actionable KPIs measure a company's ability to enact change whether through political action or a shift in company culture.
11. **Financial indicators**: Financial indicators are a marker of a business's monetary growth and stability. When paired with other KPIs, this indicator can help paint a more complete picture of your company's financial viability.
12. **Outcome indicators**: These indicators are a marker of whether the program is meeting its goals via the short or long term.

2.4.3. The advantages and disadvantages of KPIs

KPIs are often confusing and may seem abstract. However, understanding the advantages and disadvantages of performance indicators will help businesses realize the reason why they need to track them in your workplace.

2.5. Cloud Service

2.5.1. Cloud Services Definition

Cloud services are application and infrastructure resources that exist on the Internet. Third-party providers contract with subscribers for these services, allowing customers to leverage powerful computing resources without having to purchase or maintain hardware and software.

When we use cloud services, we are able to hand off managing infrastructure and focus instead on just using it. The provider we choose will support a wide range of activities that keep your business operating, such as application processing and exchange, storage, and management of our data. Using these services, our authorized users can communicate, collaborate, manage projects, and conduct data analysis, processing, sharing, and storage without needing your IT department to oversee, maintain, or back up the activity.

2.5.2. Types of Cloud Services

In each case, the service providers maintain the underlying cloud infrastructure. Other computing resources can be handled by the provider or not, as required by the subscriber's needs.

2.5.2.1. Software as a Service (SaaS)

Providers offer subscribers access to software that operates on a cloud infrastructure, enabling widespread distribution and accessibility of the application. Common types of business technologies hosted by SaaS vendors include productivity suites, customer relationship management (CRM) software, human resources management (HRM) software, and data management tools. Users can access these applications through a program interface or a thin client interface, such as a web browser. With this service, subscribers are limited to using the software; the provider manages all other aspects, including the network, servers, operating systems, storage, virtualization, data, middleware, and specific application functionalities. SaaS applications are typically designed for ease of use, catering to a broad audience.

2.5.2.2. Platform as a Service (PaaS)

PaaS (Platform as a Service) offers users more control than SaaS (Software as a Service) by providing access to a framework that starts from the operating system level. With PaaS, users can deploy their own applications on the cloud infrastructure, leveraging the programming languages, libraries, services, and tools supported by the provider. Subscribers retain control over their deployed applications, data, and possibly the configuration settings of the application-hosting environment. However, the service provider is responsible for managing the underlying network, servers, operating systems, and storage.

2.5.2.3. Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) allows subscribers to design and deploy a complete computing environment by configuring a virtual network that is isolated from other networks. Within this environment, users can install and operate their own operating systems, and provision essential computing resources such as processing power, storage, and networking components to support their software applications on the cloud infrastructure.

While IaaS users typically have control over the operating system and deployed applications, they may also have limited management capabilities over specific networking elements. In addition, many IaaS providers offer supplementary services—including system monitoring, automation tools, security features, load

balancing, and storage redundancy—to enhance performance, availability, and resilience.

2.5.2.4. Anything as a Service (XaaS)

There are several other service models defined as XaaS, which do not strictly fit in the above three categories. They are essentially anything as a service and are often limited to narrower offerings. Disaster Recovery as a Service, Communications as a Service, and Monitoring as a Service are good examples.

2.6. Tools, Methods, and Programming Languages

2.6.1 SQL Server

Microsoft SQL Server is a relational database management system (RDBMS). Applications and tools connect to a SQL Server instance or database, and communicate using Transact-SQL (T-SQL).

2.6.2 Microsoft Visual Studio

The Visual Studio IDE is a creative launching pad that you can use to edit, debug, and build code, and then publish an app. Over and above the standard editor and debugger that most IDEs provide, Visual Studio includes compilers, code completion tools, graphical designers, and many more features to enhance the software development process.

2.6.3 Multidimensional Data Analysis Technique (OLAP)

Online Analytical Processing (OLAP) is a specialized software technology that supports the multidimensional analysis of business data, enabling organizations to examine information from multiple perspectives. In modern enterprises, data is collected from a wide array of sources—including websites, mobile and desktop applications, wearable devices, and internal operational systems. OLAP systems integrate and categorize this data to provide actionable insights that inform strategic planning and data-driven decision-making.

For example, a retail organization may maintain product data—such as color, size, price, and location—in one system, while customer transaction data, including purchased items and total sales, resides in another. OLAP facilitates the integration and analysis of these disparate datasets, allowing users to answer complex analytical

queries, such as identifying the most popular product colors or evaluating the impact of product placement on sales performance.

A fundamental component of OLAP technology is the use of OLAP patterns, which act as standardized templates or best-practice frameworks for formulating and executing analytical queries. These patterns enable automation of query generation and are organized into two primary activities: pattern definition and pattern usage.

Pattern Definition involves the development of analytical templates. This process includes both a bottom-up approach, where patterns are customized within a specific data warehouse environment, and a top-down approach, where generalized patterns are refined into more specific ones using a centralized pattern catalog.

Pattern Usage pertains to the practical application of these predefined patterns. Users populate the pattern parameters with context-specific values, which allows for the execution of tailored analytical queries within the designated data environment.

Various stakeholders engage in the OLAP pattern lifecycle. Pattern authors—typically IT developers, data engineers, or data scientists with advanced expertise in business intelligence (BI) systems—are primarily responsible for defining and maintaining the patterns. Pattern users, on the other hand, include decision-makers, domain experts, business analysts, and other power users who leverage the patterns to address specific analytical needs. Importantly, individuals may alternate between these roles depending on their responsibilities and technical proficiency.

In essence, OLAP patterns serve as a mechanism for knowledge transfer, bridging the gap between technical experts and business users. They promote consistency, efficiency, and accuracy in analytical processes by encapsulating proven analytical practices and making them accessible to a wider user base within the organization.

2.6.4 Visualization Tools

Power BI is designed to seamlessly integrate with other applications in the Power Platform and the Microsoft 365 Business ecosystem. Its reports and dashboards offer a wide variety of visualizations.

Power BI can collect data from hundreds of different sources such as websites, databases, social media, and more. It allows for secure and high-integrity report publishing, enabling fast and accurate extraction of business insights.

CHAPTER 3: REQUIREMENTS ANALYSIS AND DATA MODELING

3.1. Business Requirements Analysis

3.1.1. Business Requirements

General Business Requirements of AdventureWorks Company

AdventureWorks Cycles (AWC), a multinational bicycle manufacturer headquartered in Bothell, Washington, operates in a highly competitive global bicycle market projected to reach $63.73 billion by 2025 (Statista, 2020). With approximately 300 employees and a major production facility in Mexico (Importadores Neptino), AWC faces significant challenges, including: Competition from large manufacturers, seasonal demand fluctuations, supply chain risks (material shortages and delivery delays)

Opportunities for AWC include: The fast-growing e-bike market, rising demand for sustainable transportation, emerging markets such as Vietnam (projected to hit $98.80 million in bicycle sales by 2025)

To address these challenges, AWC seeks to implement a Business Intelligence (BI) solution using AdventureWorks 2019 database, SQL Server, and Power BI. This solution will unify procurement, production, and vendor data to support data-driven decision-making via real-time interactive dashboards.

Classified	Role	What	What for / How	Why / What for
Procurement Performance	Purchasing Manager	Overview of expenses incurred, supplier reliability, and delivery performance	Displayed through a dashboard using top-cards (overview), charts, and heatmaps	Managers easily identify suppliers that need renegotiation, detect delayed orders, and monitor purchasing efficiency
Manufacturing Production	Manufacturing Manager	Production orders, completion progress, production	Use line charts, area charts, column charts, and cards to visualize	Enable timely identification of bottlenecks, allowing for adjustments to

		efficiency, number of defective products, waste costs	production status and resource utilization	production plans and optimal resource allocation
Vendor Management	Purchasing Officer	On-time delivery rate, return rate, average delivery time, product defect rate	The dashboard provides an overview using cards, bar charts, and scatter plots	Supports evaluating supplier performance and identifying partners with quality or delivery issues for renegotiation or contract non-renewal

Table 3.1. Business Requirements in Purchasing and Manufacturing Function

(Source: Authors)

3.1.2. Manufacturing Production

Main Requirements

Can the dashboard show real-time order flow, completion progress, and resource utilization? Can timeline and Gantt charts detect bottlenecks? Can gauge and bar charts identify areas needing capacity increase or maintenance?

Objectives

- Monitor production progress and resource usage.
- Detect and resolve bottlenecks.
- Optimize workforce and machine allocation, reduce downtime.

Inputs

- Production Orders: Number, status, start/end time
- Source: FactWorkOrder (via Production.WorkOrder)
- Resource Utilization: Machine & manpower usage
- Source: Production.Location
- Maintenance Data: Maintenance schedule, downtime, reasons From MES or Location data

Outputs

Top-cards (Main KPIs):

- Order Completion Rate
- Equipment Utilization Rate
- Downtime
- Cycle Time

Gantt Chart & Timeline:

- Track production progress and bottlenecks
- Gauge & Bar Charts:
- Machine usage vs. capacity
- Downtime breakdown by cause or machine

Detailed Reports:

- Order status, maintenance history, improvement suggestions
- Business Rules
- An order is “completed” if the product meets quality standards and is stored.
- Bottlenecks = step time exceeds 20% of standard time.

Equipment requires maintenance if:

- Utilization > 90% or
- Recorded incident

3.2. Data preparation

3.2.1 Overview of AdventureWorks Data

AdventureWorks is a sample database developed by Microsoft to support the exploration and study of relational database technologies within a corporate context. It simulates the operational information system of Adventure Works Cycles, a fictional large-scale manufacturer and distributor of bicycles.

The database is structured to reflect a range of core business functions, including order management, supply chain management, product management, inventory control, human resources, and financial accounting. These functional areas are organized into distinct modules, enabling users to analyze complex business processes in a realistic setting.

Unlike simplistic sample datasets that comprise only a few tables, AdventureWorks is designed as a comprehensive and practical database model. It employs multiple schemas to logically separate data by business domain, offering a more accurate representation of how enterprise-level data systems are structured and managed in real-world scenarios.

Name of Schema	Description	Total of Tables
Purchasing	Manage purchasing activities and suppliers.	6
Production	Manage products and manufacturing processes.	13
Sales	Store order and customer information.	25
HumanResources	Manage human resources and departments.	5
Person	Store contact and identity information.	11

Table 3.2. Schema in AdventureWorks database

Schema-based separation improves query efficiency while also making system maintenance simpler. Notably, AdventureWorks is ideal for developing integrated Business Intelligence (BI) systems since it supports a wide range of query types, from transactional processing (OLTP) to data analysis models (OLAP).

Data for this project was taken from AdventureWorks2019, the most recent version created to work with SQL Server 2019. This database supports multidimensional analysis and data visualization by providing the framework for building a data warehouse and creating fact and dimension tables.

An example of the AdventureWorks database's entire entity-relationship diagram (ERD) of the tables may be found below.

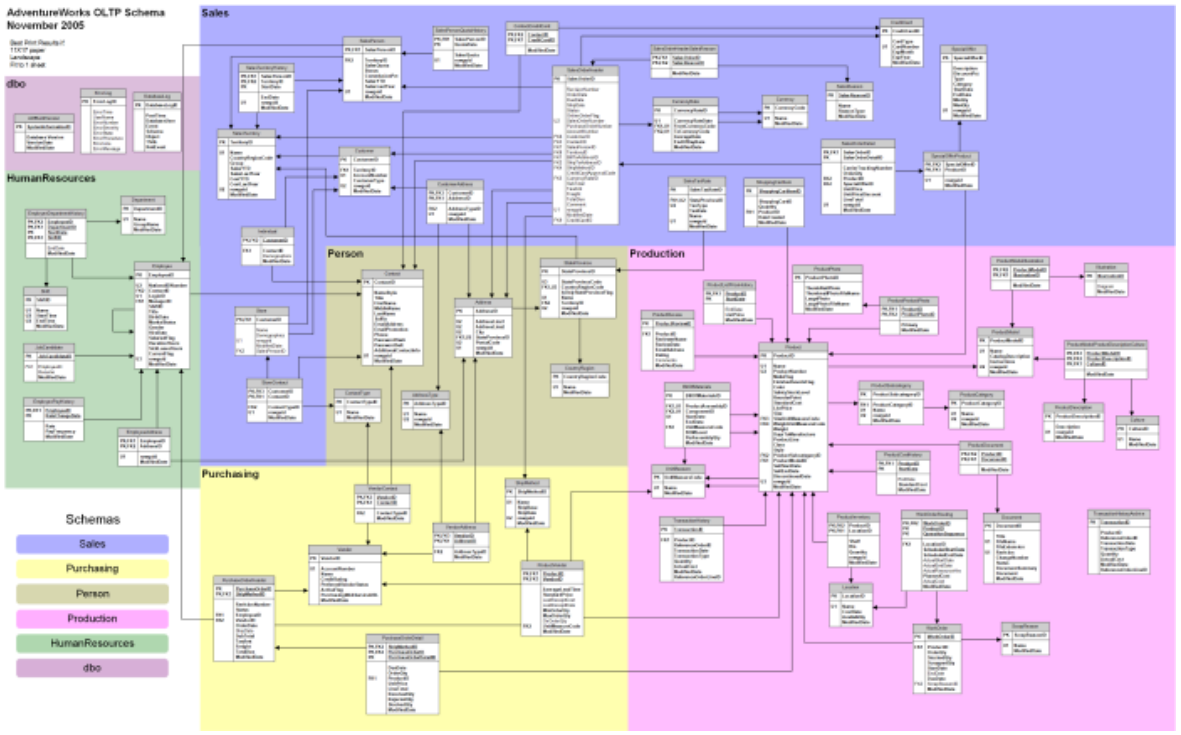


Figure 3.1. AdventureWorks ERD (Source: Improve & Repeat)

3.1.2 Data Collection

We conducted the data collection process from the AdventureWorks database to support the development of a Business Intelligence (BI) solution, ensuring that essential data points were fully extracted for business analysis, encompassing aspects related to enterprise operations and management.

The collection process was carried out through SQL queries on SQL Server. We designed queries to connect related datasets, ensuring comprehensive coverage of the aspects necessary for analysis. The extracted data was stored in temporary tables to facilitate subsequent processing and cleaning, laying the foundation for integration into the data warehouse.

3.1.3 Data Description and Data Understanding

We conducted a description and analysis of data from the AdventureWorks database, focusing on the Purchasing and Manufacturing modules, to provide detailed insights into the data supporting Business Intelligence (BI) solutions for procurement and quality management processes at AdventureWorks. The goal is to understand the data

structure, relationships, and their significance to support the analysis of cost efficiency, delivery times, and product quality, thereby optimizing business processes.

The Purchasing module records information about procurement activities from suppliers, including orders, products, costs, delivery times, and transportation methods. This data enables the analysis of procurement costs, logistics efficiency, and supplier performance. The Quality Management module focuses on product quality, including return rates, product defects, and supplier evaluations. The two modules are linked through shared fields such as product IDs and supplier IDs, supporting integrated analysis to evaluate procurement efficiency based on both cost and quality.

3.1.3.1. Module Purchasing

The Purchasing module in the AdventureWorks database is designed to collect and manage all data related to procurement, providing a comprehensive view of the supply chain process. It supports tracking activities from ordering to delivery, while also providing critical data for analyzing cost efficiency, logistics, and supplier performance. Below is the entity-relationship diagram (ERD) of the Purchasing module:

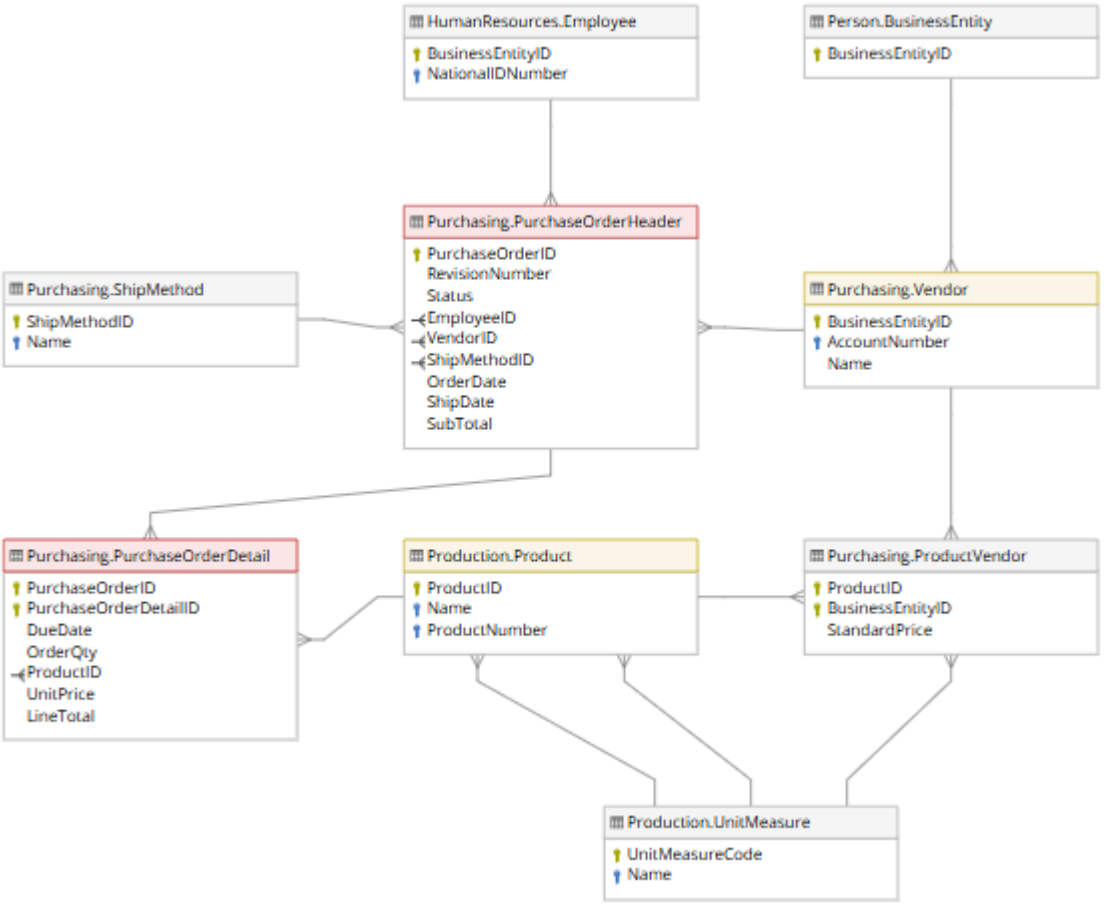


Figure 3.2. Purchasing module diagram (Source: Dataedo)

The Purchasing module consists of 5 main tables.

a. Purchasing.ProductVendor

No.	Key	Name	Data type	Description
1	PK	ProductID	int	Primary key. Foreign key to Product.ProductID.
2	PK	BusinessEntityID	int	Primary key. Foreign key to Vendor.BusinessEntityID.
3		AverageLeadTime	int	The average span of time (in days) between placing an order with the

				vendor and receiving the purchased product.
4		StandardPrice	money	The vendor's usual selling price.
5		LastReceiptCost	money	The selling price when last purchased.
6		LastReceiptDate	datetime	Date the product was last received by the vendor.
7		MinOrderQty	int	The maximum quantity that should be ordered.
8		MaxOrderQty	int	The minimum quantity that should be ordered.
9		OnOrderQty	int	The quantity currently on order.
10		UnitMeasureCode	nchar(3)	The product's unit of measure.
11		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.3. Table Purchasing.ProductVendor (Source: Dataedo)

The Purchasing.ProductVendor table plays a crucial role in managing the relationships between products and their suppliers. Key points about this table include:

- Uses a composite primary key (ProductID, BusinessEntityID) to establish links between products and the vendors supplying them, enabling many-to-many relationships.
- Contains important metrics such as AverageLeadTime, StandardPrice, LastReceiptCost, and LastReceiptDate that help evaluate vendor reliability, pricing trends, and delivery history.
- Fields like MinOrderQty, MaxOrderQty, and OnOrderQty assist in managing ordering policies by defining allowable purchase quantities and tracking current orders.

- The UnitMeasureCode field standardizes units across products, ensuring accurate inventory tracking and reporting.
- The ModifiedDate field records the latest update timestamp, supporting data integrity and audit requirements.

b. Purchasing.PurchaseOrderDetail

No.	Key	Name	Data type	Description
1	PK	PurchaseOrderID	int	Primary key. Foreign key to PurchaseOrderHeader.PurchaseOrderID.
2	PK	PurchaseOrderDetailID	int	Primary key. One line number per purchased product. Identity column.
3		DueDate	datetime	Date the product is expected to be received.
4		OrderQty	smallint	Quantity ordered.
5		ProductID	int	Product identification number. Foreign key to Product.ProductID.
6		UnitPrice	money	Vendor's selling price of a single product.
7		LineTotal	money	Per product subtotal. Computed as OrderQty * UnitPrice.
8		ReceivedQty	decimal(8,2)	Quantity actually received from the vendor.
9		RejectedQty	decimal(8,2)	Quantity rejected during inspection.

10		StockedQty	decimal(9,2)	Quantity accepted into inventory. Computed as ReceivedQty - RejectedQty.
11		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.4. Table Purchasing.PurchaseOrderDetail (Source: Dataedo)

The Purchasing.PurchaseOrderDetail table provides a detailed record of every item included within each purchase order. Key aspects of this table include:

- It uses a composite primary key (PurchaseOrderID, PurchaseOrderDetailID) to uniquely identify each product line in an order, ensuring precise tracking of individual items.
- The DueDate field specifies the expected delivery date for each product, which is essential for coordinating warehouse activities and monitoring supplier punctuality.
- Fields such as OrderQty and UnitPrice capture the quantity ordered and the vendor's price per unit, forming the basis for calculating LineTotal and thus evaluating the overall order cost.
- ReceivedQty records the actual quantity received, while RejectedQty indicates units that failed inspection. These metrics provide insight into supplier reliability and product quality.
- StockedQty, calculated as the difference between received and rejected quantities, reflects the stock successfully added to inventory, supporting accurate inventory management and minimizing discrepancies.
- The ModifiedDate field logs the last update timestamp, facilitating audit trails and maintaining the accuracy of procurement records.

c. Purchasing.PurchaseOrderHeader

No.	Key	Name	Data type	Description
1	PK	PurchaseOrderID	int	Primary key.

2		RevisionNumber	tinyint	Incremental number to track changes to the purchase order over time.
3		Status	tinyint	Order current status. 1 = Pending; 2 = Approved; 3 = Rejected; 4 = Complete.
4		EmployeeID	int	Employee who created the purchase order. Foreign key to Employee.BusinessEntityID.
5		VendorID	int	Vendor with whom the purchase order is placed. Foreign key to Vendor.BusinessEntityID.
6		ShipMethodID	int	Shipping method. Foreign key to ShipMethod.ShipMethodID.
7		OrderDate	datetime	Purchase order creation date.
8		ShipDate	datetime	Estimated shipment date from the vendor.
9		SubTotal	money	Purchase order subtotal. Computed as SUM(PurchaseOrderDetail.LineTotal) for the corresponding PurchaseOrderID.
10		TaxAmt	money	Tax amount.
11		Freight	money	Shipping cost.
12		TotalDue	money	Total due to the vendor. Computed as SubTotal + TaxAmt + Freight.
13		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.5. Table Purchasing.PurchaseOrderHeader (Source: Dataedo)

The Purchasing.PurchaseOrderHeader table captures the essential information about each purchase order issued by the company to its vendors. Key points about this table include:

- Each purchase order is uniquely identified by the PurchaseOrderID primary key, allowing precise referencing of order details.
- The RevisionNumber tracks changes to the purchase order over time, supporting order versioning and auditing.
- The Status field reflects the current state of the order, enabling workflow management—whether the order is pending, approved, rejected, or completed.
- Fields like EmployeeID and VendorID link the order to the responsible employee and the vendor receiving the order, allowing role-based tracking and performance monitoring.
- The ShipMethodID and ShipDate fields help define logistics plans and estimated delivery timelines.
- The table includes key financial fields like SubTotal, TaxAmt, Freight, and TotalDue, providing a complete cost breakdown and ensuring accurate payment calculations.
- The ModifiedDate field records when the order record was last updated, helping maintain data integrity and support for historical tracking.

d. Purchasing.ShipMethod

No.	Key	Name	Data type	Description
1	PK	ShipMethodID	int	Primary key for ShipMethod records.
2	AK	Name	nvarchar(50)	Shipping company name.
3		ShipBase	money	Minimum shipping charge.
4		ShipRate	money	Shipping charge per pound.
5	AK	rowguid	uniqueidentifier	ROWGUIDCOL number uniquely identifying the record. Used to support a merge replication sample.

6		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.6. Table Purchasing.ShipMethod (Source: Dataedo)

The Purchasing.ShipMethod table stores information about the shipping methods used in the purchasing process. Key aspects of this table include:

- The ShipMethodID field serves as the primary key, uniquely identifying each shipping method available in the system.
- The Name field stores the name of the shipping company. It functions as an alternate key to prevent duplication of shipping method names.
- The ShipBase field represents the minimum charge applied to each shipment, forming the fixed portion of the total shipping fee.
- The ShipRate field captures the cost applied per pound of goods shipped, allowing for accurate calculation of total shipping costs based on item weight.
- The rowguid field provides a globally unique identifier for each record, enabling support for merge replication and ensuring consistent synchronization across databases.
- The ModifiedDate field logs the most recent update timestamp of the record, aiding in data auditing and integrity monitoring.

e. Purchasing.Vendor

No.	Key	Name	Data type	Description
1	PK	BusinessEntityID	int	Primary key for Vendor records. Foreign key to BusinessEntity.BusinessEntityID.
2	AK	AccountNumber	nvarchar(15)	Vendor account (identification) number.
3		Name	nvarchar(50)	Company name.

4		CreditRating	tinyint	1 = Superior, 2 = Excellent, 3 = Above average, 4 = Average, 5 = Below average.
5		PreferredVendorStatus	bit	0 = Do not use if another vendor is available. 1 = Preferred over other vendors supplying the same product.
6		ActiveFlag	bit	0 = Vendor no longer used. 1 = Vendor is actively used.
7		PurchasingWebServiceURL	nvarchar(1024)	Vendor URL.
8		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.7. Table Purchasing.Vendor (Source: Dataedo)

The Purchasing.Vendor table maintains detailed information about the vendors that supply products to the organization. Key aspects of this table include:

- The BusinessEntityID field uniquely identifies each vendor and also serves as a foreign key linking to the BusinessEntity table, enabling unified tracking of all business entities.
- The AccountNumber field records the vendor's identification number used for accounting, invoicing, and communication purposes.
- The Name field stores the vendor's official company name for reference in procurement documents and reports.
- The CreditRating field categorizes vendors into five performance levels, from Superior to Below Average, assisting in risk assessment and sourcing decisions.
- The PreferredVendorStatus bit field indicates whether a vendor is prioritized over others when sourcing the same product, supporting strategic vendor relationships.

- The ActiveFlag field shows whether a vendor is currently in use, enabling the company to filter out inactive or outdated vendor records.
- The PurchasingWebServiceURL field provides a URL for web service integration, enabling electronic procurement and communication with vendors.
- The ModifiedDate field logs the most recent update to the vendor record, supporting data integrity and audit requirements.

3.1.3.2. Module Manufacturing

The Manufacturing module in the AdventureWorks database is designed to collect and manage all data related to the manufacturing process, providing a comprehensive view of production activities. It supports tracking processes from planning to product completion, while also providing critical data for analyzing production efficiency, costs, and quality. Below is the entity-relationship diagram (ERD) of the Manufacturing module:

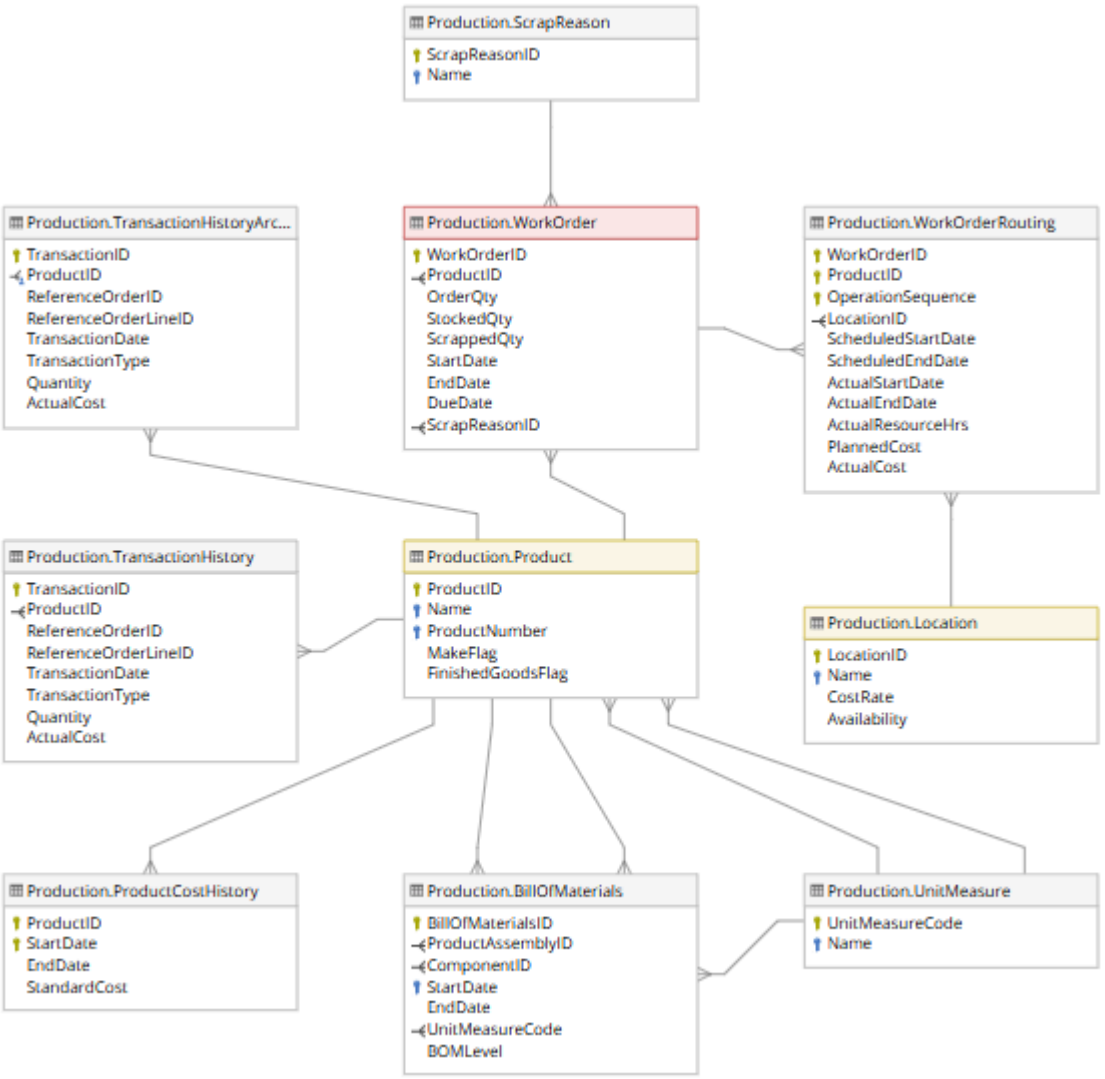


Figure 3.3. Manufacturing module diagram (Source: Dataedo)

The Manufacturing module consists of 10 tables.

a. Production.BillOfMaterials

No.	Key	Name	Data type	Description
1	PK	BillOfMaterialsID	int	Primary key for BillOfMaterials records.

2	AK	ProductAssemblyID	int	Parent product identification number. Foreign key to Product.ProductID.
3	AK	ComponentID	int	Component identification number. Foreign key to Product.ProductID.
4	AK	StartDate	datetime	Date the component started being used in the assembly item.
5		EndDate	datetime	Date the component stopped being used in the assembly item.
6		UnitMeasureCode	nchar(3)	Standard code identifying the unit of measure for the quantity.
7		BOMLevel	smallint	Indicates the depth the component is from its parent (AssemblyID).
8		PerAssemblyQty	decimal(8,2)	Quantity of the component needed to create the assembly
9		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.8. Table Production.BillOfMaterials (Source: Dataedo)

The Production.BillOfMaterials table defines the hierarchical structure of manufactured products, specifying which components are required to build a parent product (assembly). This table is central to the manufacturing process and supports planning, costing, and inventory control.

- Each row links a parent product (ProductAssemblyID) to one of its components (ComponentID), forming a multi-level assembly structure.
- BOMLevel field defines the depth of the component in the hierarchy, allowing for analysis of complex assemblies and subassemblies.

- The PerAssemblyQty field specifies the exact amount of the component needed per unit of the final product, which is essential for material requirement planning (MRP).
- The validity period of each component (from StartDate to EndDate) allows manufacturers to track and adapt to changes in product design over time.
- Combined with the Product, UnitMeasure, and WorkOrder tables, this data supports full traceability and efficiency in the production planning process.

b. Production.WorkOrder

No.	Key	Name	Data type	Description
1	PK	WorkOrderID	int	Primary key for WorkOrder records.
2		ProductID	int	Product identification number. Foreign key to Product.ProductID.
3		OrderQty	int	Product quantity to build.
4		StockedQty	int	Computed: isnull([OrderQty] - [ScrappedQty], 0). Quantity built and put into inventory.
5		ScrappedQty	smallint	Quantity that failed inspection.
6		StartDate	datetime	Work order start date.
7		EndDate	datetime	Work order end date.
8		DueDate	datetime	Work order due date.
9		ScrapReasonID	smallint	Reason for inspection failure. Foreign key to Production.ScrapReason.
10		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.9. Table Production.WorkOrder (Source: Dataedo)

The Production.WorkOrder table records detailed information about each manufacturing work order, tracking the production of products through various stages. This table is critical for managing and monitoring production workflows, inventory levels, and quality control processes. Key features include:

- The table logs the quantity of products to be built (OrderQty), quantities successfully completed and stocked (StockedQty), and quantities scrapped due to defects (ScrappedQty).
- Fields such as StartDate, EndDate, and DueDate help manage and track the timing of production activities to ensure on-time completion.
- The ScrapReasonID field links to the Production.ScrapReason table to specify reasons for any scrapped items, aiding root cause analysis and process improvements.
- The ProductID foreign key connects each work order to the specific product being manufactured, enabling inventory and product lifecycle management.
- The ModifiedDate records the last update timestamp, ensuring changes to work order records are tracked for accuracy and accountability.

c. Production.WorkOrderRouting

No.	Key	Name	Data type	Description
1	PK	WorkOrderID	int	Primary key. Foreign key to Production.WorkOrder.WorkOrderID.
2	PK	ProductID	int	Primary key. Foreign key to Production.Product.ProductID.
3	PK	OperationSequence	smallint	Primary key. Indicates the manufacturing process sequence.

4		LocationID	smallint	Manufacturing location where the part is processed. Foreign key to Production.Location.LocationID.
5		ScheduledStartDate	datetime	Planned manufacturing start date.
6		ScheduledEndDate	datetime	Planned manufacturing end date.
7		ActualStartDate	datetime	Actual start date.
8		ActualEndDate	datetime	Actual end date.
9		ActualResourceHrs	decimal(9,4)	Number of manufacturing hours used.
10		PlannedCost	money	Estimated manufacturing cost.
11		ActualCost	money	Actual manufacturing cost.
12		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.10. Table Production.WorkOrderRouting (Source: Dataedo)

The Production.WorkOrderRouting table manages detailed information about the routing of work orders through the manufacturing process. It tracks the sequence of operations required to complete production and captures both planned and actual timings and costs. Key functions of this table include:

- The composite primary key (WorkOrderID, ProductID, and OperationSequence) defines the ordered steps a product must undergo during manufacturing.
- The LocationID specifies the facility or workstation where each operation takes place, supporting resource allocation and capacity planning.
- Fields such as ScheduledStartDate and ScheduledEndDate record planned timelines, while ActualStartDate and ActualEndDate capture real execution times, enabling variance analysis.

- ActualResourceHrs and cost fields (PlannedCost, ActualCost) provide insight into labor usage and financial performance per operation.
- The ModifiedDate tracks the last update, ensuring the integrity and traceability of routing information over time.

e. Production.ScrapReason

No.	Key	Name	Data type	Description
1	PK	ScrapReasonID	smallint	Primary key for ScrapReason records.
2	AK	Name	nvarchar(50)	Failure description.
3		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.11. Table Production.ScrapReason (Source: Dataedo)

The Production.ScrapReason table stores a list of standardized reasons for scrapping or rejecting items during the production or procurement process. This table is essential for quality tracking and root cause analysis of product defects. Key aspects of this table include:

- The Name field provides a textual description of the reason why a product was rejected or scrapped, such as “Damaged packaging” or “Defective component.” This helps standardize failure reporting across the organization.
- This table is commonly linked to the WorkOrder table through the ScrapReasonID to provide context for the quantity of scrapped items in manufacturing.
- The ModifiedDate field records the most recent time the reason entry was updated, which is important for maintaining a consistent and auditable list of scrap causes.
- By linking defect occurrences to specific reasons, the company can identify frequent issues, prioritize corrective actions, and improve supplier or process quality over time.

f. Production.TransactionHistoryArchive

No.	Key	Name	Data type	Description
1	PK	TransactionID	int	Primary key for TransactionHistoryArchive records.
2		ProductID	int	Product identification number. Foreign key to Production.ProductID.
3		ReferenceOrderID	int	Purchase order, sales order, or work order identification number.
4		ReferenceOrderLineID	int	Line number associated with the purchase order, sales order, or work order.
5		TransactionDate	datetime	Date and time of the transaction.
6		TransactionType	nchar(1)	W = Work Order, S = Sales Order, P = Purchase Order.
7		Quantity	int	Product quantity.
8		ActualCost	money	Product cost.
9		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.12. Table Production.TransactionHistoryArchive (Source: Dataedo)

The Production.TransactionHistoryArchive table maintains detailed information about historical transactions related to products within the organization. Key aspects of this table include:

- The TransactionID field uniquely identifies each transaction record and serves as the primary key.
- The ProductID field links to the Production.Product table, enabling tracking of product-specific transactions.
- The ReferenceOrderID field records the identification number of the associated purchase order, sales order, or work order.
- The ReferenceOrderLineID field specifies the line number tied to the respective order, aiding in detailed transaction tracking.
- The TransactionDate field logs the date and time when the transaction occurred, supporting temporal analysis.
- The TransactionType field categorizes the transaction type (Work Order, Sales Order, or Purchase Order), facilitating process differentiation.
- The Quantity field indicates the number of products involved in the transaction, useful for inventory management.
- The ActualCost field records the actual cost of the product, supporting cost analysis and budgeting.
- The ModifiedDate field tracks the most recent update to the transaction record, ensuring data accuracy and auditability.

g. Production.TransactionHistory

No.	Key	Name	Data type	Description
1	PK	TransactionID	int	Primary key for TransactionHistory records.
2		ProductID	int	Product identification number. Foreign key to Product.ProductID.
3		ReferenceOrderID	int	Purchase order, sales order, or work order identification number.

4		ReferenceOrderLineID	int	Line number associated with the purchase order, sales order, or work order.
5		TransactionDate	datetime	Date and time of the transaction.
6		TransactionType	nchar(1)	W = WorkOrder, S = SalesOrder, P = PurchaseOrder.
7		Quantity	int	Product quantity.
8		ActualCost	money	Product cost.
9		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.13. Table Production.TransactionHistory (Source: Dataedo)

The TransactionHistory table is essential for tracking transactional data related to products. Key points about this table include:

- Utilizes TransactionID as the primary key to uniquely identify each transaction record.
- Links to ProductID to associate transactions with specific products, enabling detailed product transaction analysis.
- Includes ReferenceOrderID and ReferenceOrderLineID to track orders and their line items across purchase, sales, or work orders.
- The TransactionDate and TransactionType fields provide temporal and categorical context to each transaction.
- Fields like Quantity and ActualCost support inventory and cost management by recording product amounts and associated costs.
- The ModifiedDate field ensures data integrity by logging the latest update timestamp for auditing purposes.

h. Production.ProductCostHistory

No.	Key	Name	Data type	Description
1	PK	ProductID	int	Product identification number. Foreign key to Product.ProductID.
2	PK	StartDate	datetime	Product cost start date.
3		EndDate	datetime	Product cost end date.
4		StandardCost	money	Standard cost of the product.
5		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.14. Table Production.ProductCostHistory (Source: Dataedo)

The ProductCostHistory table is vital for tracking the cost history of products over time. Key points about this table include:

- Uses ProductID as the primary key to uniquely identify each product cost record, linking to Product.ProductID for product association.
- The StartDate and EndDate fields define the time period for which the standard cost is applicable.
- The StandardCost field records the standard cost of the product during the specified period.
- The ModifiedDate field logs the latest update timestamp, ensuring data integrity and supporting audit requirements.

i. Production.Location

No.	Key	Name	Data type	Description
1	PK	LocationID	smallint	Primary key for Location records.
2	AK	Name	nvarchar(50)	Location description.

3		CostRate	smallmoney	Standard hourly cost of the manufacturing location.
4		Availability	decimal(8,2)	Work capacity (in hours) of the manufacturing location.
5		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.15. Table Production.Location (Source: Dataedo)

The Location table is crucial for managing information about manufacturing locations. Key points about this table include:

- Uses LocationID as the primary key to uniquely identify each location record.
- The Name field provides a descriptive label for the manufacturing location.
- The CostRate field records the standard hourly cost associated with the location.
- The Availability field tracks the work capacity in hours for the manufacturing location.
- The ModifiedDate field logs the latest update timestamp, ensuring data integrity and supporting audit requirements.

3.1.3.3. Related Tables

a. Production.Product

No.	Key	Name	Data type	Description
1	PK	ProductID	int	Primary key for Product records.
2	AK	Name	nvarchar(50)	Name of the product.
3	AK	ProductNumber	nvarchar(25)	Unique product identification number.

4		MakeFlag	bit	0 = Product is purchased, 1 = Product is manufactured in-house. Default: 1.
5		FinishedGoodsFlag	bit	0 = Product is not a salable item. 1 = Product is salable. Default: 1.
6		Color	nvarchar(15)	Product color.
7		SafetyStockLevel	smallint	Minimum inventory quantity.
8		ReorderPoint	smallint	Inventory level that triggers a purchase or work order.
9		StandardCost	money	Standard cost of the product.
10		ListPrice	money	Selling price.
11		Size	nvarchar(5)	Product size.
12	FK	SizeUnitMeasureCode	nchar(3)	Foreign key to Production.UnitMeasure. Unit of measure for Size.
13	FK	WeightUnitMeasureCode	nchar(3)	Foreign key to Production.UnitMeasure. Unit of measure for Weight.
14		Weight	decimal(8,2)	Product weight.
15		DaysToManufacture	int	Number of days required to manufacture the product.
16		ProductLine	nchar(2)	R = Road, M = Mountain, T = Touring, S = Standard.

17		Class	nchar(2)	H = High, M = Medium, L = Low.
18		Style	nchar(2)	W = Womens, M = Mens, U = Universal.
19	FK	ProductSubcategoryID	int	Foreign key to Production.ProductSubcategory.
20	FK	ProductModelID	int	Foreign key to Production.ProductModel.
21		SellStartDate	datetime	Date the product was available for sale.
22		SellEndDate	datetime	Date the product was no longer available for sale.
23		DiscontinuedDate	datetime	Date the product was discontinued.
24		rowguid	uniqueidentifier	Unique identifier. Used for merge replication.
25		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.16 Table Production.Product (Source: Dataedo)

The Production.Product table stores detailed information about products managed within the manufacturing system.

- It includes both finished goods available for sale and components used in assemblies.
- Fields such as MakeFlag and FinishedGoodsFlag distinguish whether a product is manufactured in-house or purchased, and if it is salable.

- Inventory-related fields like SafetyStockLevel and ReorderPoint support stock control and procurement planning.
- Pricing and cost details are stored in StandardCost and ListPrice.
- Product attributes such as Color, Size, Weight, and classification fields (ProductLine, Class, Style) help categorize and describe products.
- The table links to product subcategories and models through foreign keys, enabling detailed product hierarchy.
- Date fields track the availability and lifecycle of products.
- Unique identifiers ensure data integrity and support replication.

b. Production.UnitMeasure

No.	Key	Name	Data type	Description
1	PK	UnitMeasureCode	nchar(3)	Primary key.
2		Name	nvarchar(50)	Unit of measure description.
3		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.17. Table Production.UnitMeasure (Source: Dataedo)

The UnitMeasure table is essential for defining and managing units of measure used in the system. Key points about this table include:

- Uses UnitMeasureCode as the primary key to uniquely identify each unit of measure.
- The Name field provides a descriptive name for the unit of measure.
- The ModifiedDate field logs the latest update timestamp, ensuring data integrity and supporting audit requirements.

c. HumanResources.Employee

No.	Key	Name	Data type	Description
1	PK	BusinessEntity ID	int	Primary key for Employee records. Foreign key to Person.BusinessEntityID.

2		NationalIDNumber	nvarchar(15)	Unique national identification number such as a social security number.
3		LoginID	nvarchar(256)	Network login. Test2.
4		OrganizationNode	hierarchyid	Where the employee is located in the corporate hierarchy.
5		OrganizationLevel	smallint	The depth of the employee in the corporate hierarchy.
6		JobTitle	nvarchar(50)	Work title such as Buyer or Sales Representative.
7		BirthDate	date	Date of birth.
8		MaritalStatus	nchar(1)	M = Married, S = Single.
9		Gender	nchar(1)	M = Male, F = Female.
10		HireDate	date	Employee hired on this date.
11		SalariedFlag	bit	Job classification: 0 = Hourly, not exempt from collective bargaining; 1 = Salaried, exempt from collective bargaining.
12		VacationHours	smallint	Number of available vacation hours.
13		SickLeaveHours	smallint	Number of available sick leave hours.
14		CurrentFlag	bit	0 = Inactive, 1 = Active.
15		rowguid	uniqueidentifier	ROWGUIDCOL number uniquely identifying the record. Used to support a merge replication sample.

16		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.18. Table HumanResources.Employee (Source: Dataedo)

The Employee table is critical for managing employee-related data within the organization. Key points about this table include:

- Uses BusinessEntityID as the primary key to uniquely identify each employee, linking to Person.BusinessEntityID.
- The NationalIDNumber and LoginID fields provide unique identification and login credentials for employees.
- OrganizationNode and OrganizationLevel define the employee's position within the corporate hierarchy.
- Fields like JobTitle, BirthDate, MaritalStatus, Gender, and HireDate capture personal and employment details.
- SalariedFlag, VacationHours, SickLeaveHours, and CurrentFlag support job classification and leave management.
- The rowguid and ModifiedDate fields ensure unique record identification and track the latest updates for audit purposes.

d. Person.BusinessEntity

No.	**Key**	**Name**	**Data type**	**Description**
1	PK	BusinessEntityID	int	Primary key for all customers, vendors, and employees.
2		rowguid	uniqueidentifier	ROWGUIDCOL number uniquely identifying the record. Used to support a merge replication sample.
3		ModifiedDate	datetime	Date and time the record was last updated.

Table 3.19. Table Person.BusinessEntity (Source: Dataedo)

The Person.BusinessEntity table is fundamental for managing unique identifiers across customers, vendors, and employees. Key points about this table include:

- Uses BusinessEntityID as the primary key to uniquely identify records for customers, vendors, and employees.
- The rowguid field provides a unique identifier for each record, supporting merge replication processes.
- The ModifiedDate field logs the latest update timestamp, ensuring data integrity and audit capabilities.

3.1.4 EDA and data cleaning

3.1.4.1. EDA

The chart below shows the changes in purchasing costs of AdventureWorks from 2011 to 2014. In 2011, total purchasing costs started at a very low level, nearly zero. By 2012, costs significantly increased to around 4 million USD, reflecting the beginning of expanded purchasing activities. The year 2013 marked strong growth, with costs rising to approximately 20 million USD. The peak was recorded in 2014, when total purchasing costs reached about 40 million USD, indicating a significant increase in purchasing demand or scale of operations.

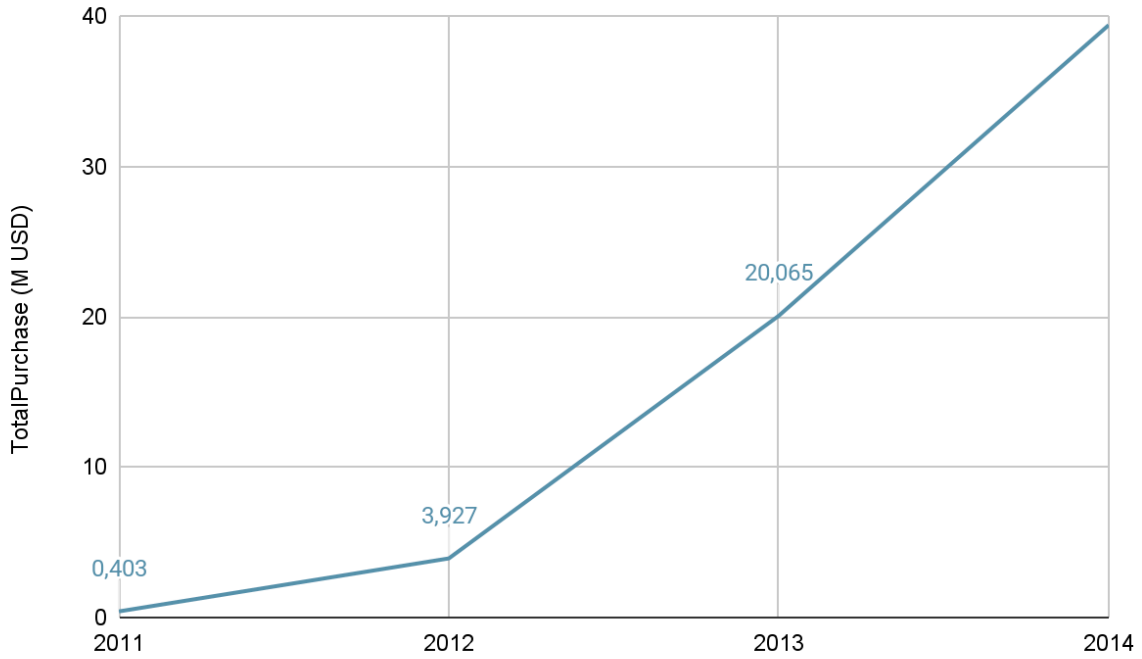


Figure 3.4. Fluctuation of purchasing costs over time

The diagram illustrates the total production costs by manufacturing areas of AdventureWorks, with units in trillion USD. The Frame Forming area leads with the highest costs, reaching 685.391 trillion USD, indicating that this area has the largest production activity or labor costs. Following are Frame Welding with 292.218 trillion USD and Final Assembly with 285.984 trillion USD, reflecting the crucial roles of machining and final assembly stages. Areas such as Deburr and Polish, Paint, and Specialized Paint have significantly lower costs, at 24.379, 15.961, and 8.015 trillion USD, respectively, indicating lower levels of investment or output in these stages.

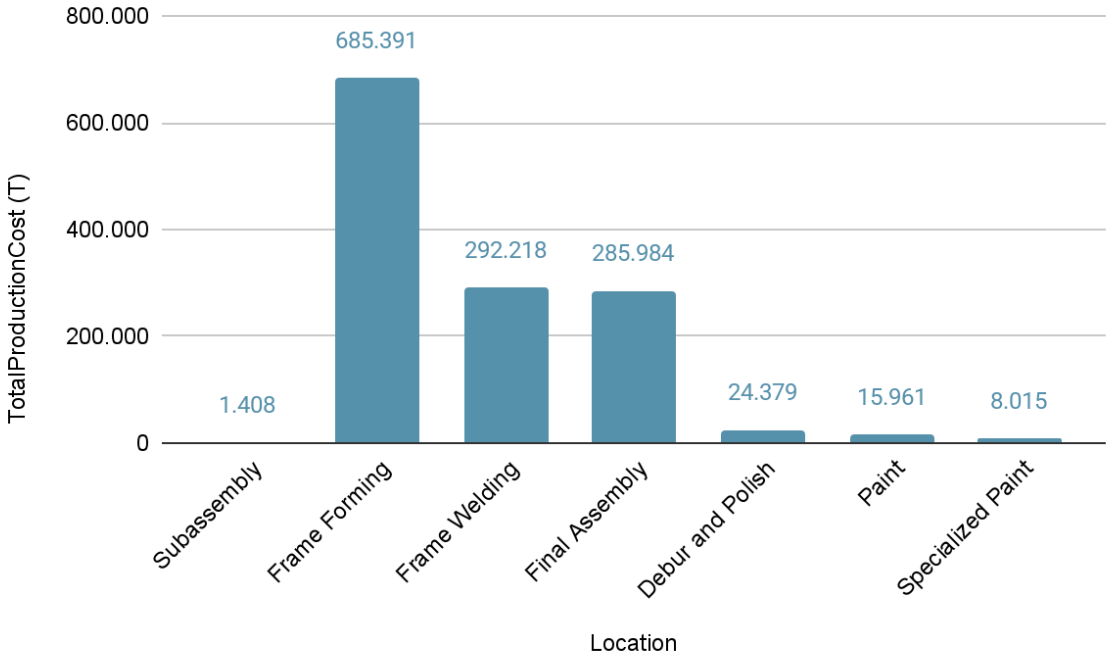


Figure 3.5. The total cost by manufacturing region (Source: Authors)

3.1.4.2. Data cleaning

After gaining a comprehensive understanding of the data within the Purchasing and Manufacturing modules of the AdventureWorks database, a detailed analysis was conducted on the columns of the relevant datasets. Exploratory Data Analysis (EDA) was performed on the Purchasing.ProductVendor, Purchasing.PurchaseOrderDetail, Purchasing.PurchaseOrderHeader, Purchasing.ShipMethod, Purchasing.Vendor, Production.ScrapReason, Production.WorkOrder, Production.ProductReview, and Production.WorkOrderRouting tables to identify data quality issues, such as null values and outliers, and to prepare the data for Business Intelligence (BI) analysis.

Null Ratio

The results of checking for null values reveal that several tables contain a significant proportion of empty values. In many cases, the null ratio exceeds 40%, necessitating a careful evaluation based on the importance of each column for the Purchasing and Manufacturing processes, as well as BI requirements. The columns below, which have a high null value ratio and are deemed non-critical for analysis, will be excluded as identified in the table below.

No.	Table Name	Feature	Null Ratio
1	Purchasing.Vendor	PurchasingWebServiceURL	94.23%
2	Production.Product	Color	49.21%
3	Production.Product	Size	58.13%
4	Production.Product	Weight	59.33%
5	Production.Product	ProductLine	44.84%
6	Production.Product	Class	51%
7	Production.Product	Style	58.13%
8	Production.Product	DiscontinuedDate	100%
9	Production.Product	SellEndDate	88.56%
10	Production.Product	ProductModelID	41.47%
11	Production.Product	ProductSubcategoryID	41.47%
12	Production.Product	WeightUnitMeasureCode	59.33%
13	Production.Product	SizeUnitMeasureCode	65%
14	Production.WorkOrder	ScrapReasonID	99%
15	Production.ProductCostHistory	EndDate	49.37%
16	Production.BillOfMaterials	EndDate	92.57%
17	Purchasing.ProductVendor	OnOrderQty	66.3%

Table 3.20. Null Ratio

Additionally, the Production.Product table, integral to the Purchasing and Manufacturing processes, was identified to contain numerous columns with a high

percentage of empty values. The figure below illustrates that some of these columns have a null value ratio exceeding 50%. In this table, features with significant empty values are excluded, except for ProductSubcategoryName and ProductLine, which are preserved due to their critical role in product categorization and KPI calculations, notwithstanding their association with elevated null ratios in related datasets.

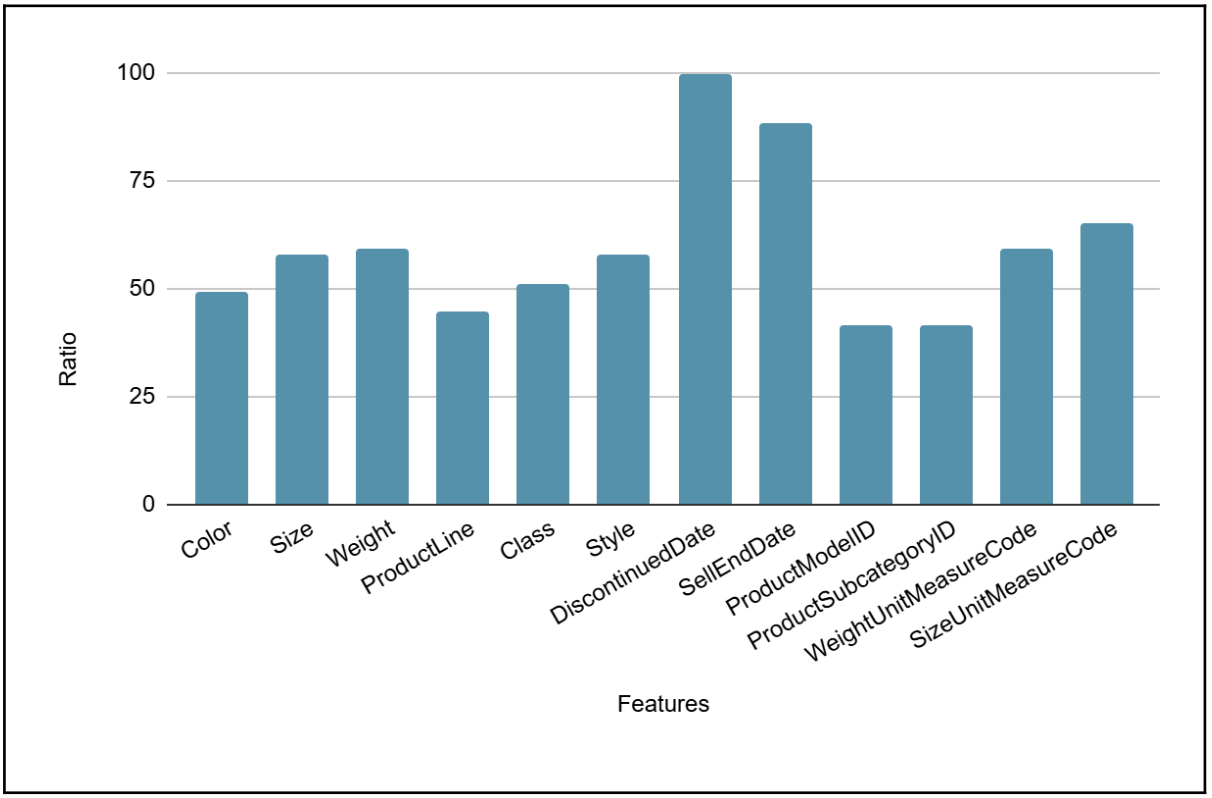


Figure 3.6. The null value ratio of the Production.Product table (Source: Authors)

3.2 Data Warehouse Design

Since the Purchase and Work Order modules are essential to the manufacturing process and cover both input and output procedures, we have chosen them as our two main modules. Purchasing raw materials from vendors is reflected in the Purchase module. It offers useful information for tracking purchase order history, delivery schedules, cost control, supplier performance analysis, and inventory management optimization. This module is essential to the production process since it guarantees the timely availability of resources.

The Work Order module, on the other hand, includes comprehensive details about manufacturing processes, such as the materials utilized, production schedules, output

levels, and the different stages that are involved. By analyzing this data, production efficiency, bottleneck identification, quality control, and product traceability can all be assessed. Notably, there is a strong relationship between the two modules: materials procured through the Purchase module are directly consumed in the Work Orders.

Therefore, choosing these two modules allows for a comprehensive view of the core activities in a manufacturing business and lays a solid foundation for conducting in-depth analyses related to cost, operational performance, and product quality.

3.2.1. Business Processes

3.2.1.1. Procurement Performance

The purchasing manager needs a quick overview of total spending, vendor reliability, and delivery efficiency right upon opening the dashboard. Key metrics displayed in top-level cards provide a real-time snapshot of procurement performance, helping managers quickly assess overall status and trends.

Trend charts and heatmaps are used to highlight underperforming vendors—those with delayed deliveries, low reliability, or overspending. With this insight, the purchasing manager can proactively plan for contract renegotiations or vendor switching to improve cost control and delivery timelines.

3.2.1.2. Vendor Management

The procurement officer needs clear visibility into which vendors deliver on time, which tend to delay, and which supply defective products. Upon opening the dashboard, users are presented with summary indicators such as on-time delivery rate, return rate, and average delivery time, offering an objective evaluation of vendor performance.

Bar charts and scatter plots help easily identify high-risk or underperforming vendors, enabling the team to flag them for renegotiation or potential replacement. Monitoring these trends over time helps improve the supply chain's overall quality and ensures consistent service levels.

3.2.1.3. Manufacturing Production

The production manager wants visibility into work order volume, completion progress, and resource utilization to make timely adjustments in manpower or

machinery. The dashboard provides a comprehensive view of the production flow with intuitive and interactive data visualization.

Tools like timeline charts and Gantt charts help detect bottlenecks early, allowing for efficient reallocation of tasks. Gauge charts and bar charts serve as alerts for areas where capacity needs to be increased or equipment maintenance is required to prevent disruptions in the production line.

3.2.2. Bus Matrix

Business Process	**Product**	**Vendor**	**Ship Method**	**ScrapReason**	**Time**
Procurement Performance	X	X	X		X
Vendor Management	X	X	X		X
Manufacturing Production	X			X	X

Table 3.21. Bus Matrix

3.2.3. Master Data

We realized that the master data primarily consists of vendors currently working with us, reflecting a strong focus on a B2B (Business-to-Business) model.

3.3. Facts and Dimesions

3.3.1 Data mapping

3.3.2 Dimensions tables

AdventureWork2019				DataWarehouse				ETLResult
Table	Attribute Name	Data type	Is Null	Attribute Name	Data type	Is Null	Description	
Purchasing.ShipMethod				DimShipMethod				

				ShipMethodKey (PK)	int		Primary Key for Ship Method Dimension	Auto Generate
Purchasing.ShipMethod	ShipMethodID	int		ShipMethodID	int		Natural Key for ShipMethod Dimension	From Source
	Name	nvarchar(50)		ShipName	nvarchar(50)		Name of the shipping company	
				StartDate	date		Start day of ShipMethod	Auto Generate
				EndDate	date	x	End day of ShipMethod	
Purchasing.Vendor Purchasing.vVendorWithAddresses				DimVendor				
				VendorKey	int		Primary Key for Vendor Dimension	Auto Generate
Purchasing.Vendor	BusinessEntityID	int		VendorID	int		Natural Key for Vendor Dimension	From Source
	Name	nvarchar(50)		VendorName	nvarchar(50)		Name of the vendor or supplier	
	ActiveFlag	bit		ActiveFlag	bit		Indicates whether the vendor is currently active (1 = Active, 0 = Inactive)	
	PreferredVendorStatus	bit		PreferredVendorStatus	bit		Indicates if this is a preferred vendor (1 = Preferred, 0 = Not Preferred)	

	CreditRating	tinyint		CreditRating	tinyint		A numeric rating representing the vendor's creditworthiness	
Purchasing.vVendorWithAddresses	City	nvarchar(30)		CityName	nvarchar(30)	x	Name of the city where the vendor is located	
Purchasing.vVendorWithAddresses	CountryRegionName	nvarchar(50)		CountryRegionName	nvarchar(50)	x	Name of the country or region where the vendor is located	
				StartDate	date		Start day of Vendor	Auto Generate
				EndDate	date	x	End day of Vendor	
Production.ScrapReason				DimScrapReason				
				SceapReasonID	INT		Primary Key of the ScrapReason Dimension	From Source
Production.WorkOrderRouting				ScrapReasonName	nvarchar(50)		Name of scrap reason	
				StartTime	datetime		Start day of ScrapReason	Auto Generate
				EndTime	datetime	x	End day of ScrapReason	
Production.Product Production.ProductSubcategory Production.ProductCategory				DimProduct				
				ProductKey	int		Primary Key of the Product Dimension	Auto Generate

Production.Product	ProductID	int		ProductID	int		Natural Key for Product Dimension	From Source
	Name	nvarchar(50)		ProductName	nvarchar(50)		Name of the product	
	ProductNumber	nvarchar(25)		ProductNumber	nvarchar(25)		Internal product number used for identification	
	MakeFlag	bit		MakeFlag	bit		Indicates if the product is manufactured (1) or purchased (0)	
	FinishedGoodsFlag	bit		FinishedGoodsFlag	bit		Indicates if the product is a finished good (1 = Yes, 0 = No)	
	SafetyStockLevel	smallint		SafetyStockLevel	smallint		Minimum inventory level to avoid stockouts	
	ReorderPoint	smallint		ReorderPoint	smallint		Inventory level that triggers a reorder action	
	DaysToManufacture	int		DaysToManufacture	int		Number of days required to manufacture the product	
	ProductLine	nchar(2)		ProductLine	nchar(2)	x	Code representing the product line (e.g., R = Road, M = Mountain)	

Production.ProductSubcategory	Name	nvarchar(50)		ProductSubcategoryName	nvarchar(50)	x	Name of the product subcategory	
Production.ProductCategory	Name	nvarchar(50)		ProductCategoryName	nvarchar(50)	x	Name of the product category	
				StartDate	date		Start day of Product	Auto Generate
				EndDate	date	x	End day of Product	
				DimTime				
				DateKey	int		Primary Key of the Time Dimension	
				FullTime	datetime		Full timestamp (date + time)	
				TheDay	tinyint	x	Numeric day of the month (1–31)	
				TheDayName	nvarchar(30)		Name of the day (e.g., Monday, Tuesday)	
				TheWeek	tinyint	x	Week number within the month or year	
				TheISOWeek	tinyint	x	Week number according to ISO standard (1–53)	
				TheDayofWeek	tinyint		Numeric value of the day in the week (1 = Monday, 7 = Sunday)	
				TheMonth	tinyint	x	Numeric month (1 = January, 12 = December)	

SQL Script

				TheMonthName	nvarchar(30)	x	Full name of the month (e.g., January)	
				TheQuarter	tinyint	x	Quarter of the year (1–4)	
				TheYear	smallint	x	Year in four digits	
				TheFirstOfMonth	date	x	First date of the current month	
				TheLastOfYear	date	x	Last date of the current year	
				TheDayofYear	smallint	x	Day number within the year (1–366)	

Table 3.22. ETL Mapping Dimension Tables (Source: Authors)

3.3.3 Fact tables

AdventureWork2019				DataWarehouse				ETLResult
Table	Attribute Name	Data type	Is Null	Attribute Name	Data type	Is Null	Description	
Purchasing.PurchaseOrderHeader Purchasing.PurchaseOrderDetail				FactPurchaseOrder				
				ProductKey	int	x	Natural key for PurchaseOrder record	From DimProduct
				VendorKey	int	x	Natural key for PurchaseOrder record	From DimVendor
				ShipMethodKey	int	x	Natural key for PurchaseOrder record	From DimShipMethod

Purchasing.PurchaseOrderHeader	PurchaseOrderID	int		PurchaseOrderID	int	x	Unique number for purchase order	
	OrderDate	datetime		OrderDate	datetime	x	Date the purchase order was created	
	ShipDate	datetime		ShipDate	datetime	x	Date the products were shipped by the vendor	
	Freight	money		Freight	money	x	Freight cost associated with the purchase order	
	Status	tinyint		Status	tinyint	x	Status of the purchase order (e.g., 1 = Pending, 2 = Approved)	
Purchasing.PurchaseOrderDetail	PurchaseOrderDetailID	int		PurchaseOrderDetailID	int	x	Line-level identifier of the purchase order	From Source
	DueDate	datetime		DueDate	datetime	x	Date the product is expected to be delivered	
	QuantityOrdered	smallint		QuantityOrdered	smallint	x	Number of units ordered for the product	
	QuantityReceived	decimal(8, 2)		QuantityReceived	decimal(8, 2)	x	Unit price agreed upon for the product	
	UnitPrice	money		UnitPrice	money	x	Number of units actually received from the vendor	
	LineTotal	money		LineTotal	money	x	Total cost of the line	

							item	
				QuantityRejected	decimal(8, 2)	x	Number of units rejected upon receipt	
				IsDelayed	bit	x	Indicates whether the delivery was delayed (1 = Yes, 0 = No)	
				IsCompleted	bit	x	Indicates whether the purchase order line is fully received (1 = Yes, 0 = No)	SQL Script
Production.WorkOrder, Production.WorkOrderRouting				FactWorkOrder				
				ProductKey	int	x	Natural key for WorkOrder record	From DimProduct
Production.WorkOrder	WorkOrderID	int		WorkOrderID	int	x	Unique number for work order	From Source
	StartDate	datetime		StartDate	datetime	x	Actual start date of the work order	
	EndDate	datetime		EndDate	datetime	x	Actual end date of the work order	
	OrderQty	int		OrderQty	int	x	Number of units to be produced	
	ScrappedQty	int		ScrappedQty	int	x	Number of units scrapped during production	
	DueDate	datetime		DueDate	datetime	x	Work order due date	
	ScrapReason	Smallint		ScrapReason	Smallint	x	Reason for inspection	

	ID			ID			failure	
Production.WorkOrderRouting	ActualCost	money		ActualCost	money	x	Total actual cost incurred during production	
	ActualResourceHrs	decimal(9, 4)		ActualResourceHrs	decimal(9, 4)	x	Actual labor or machine hours consumed during the process	
	ScheduledStartDate	datetime		ScheduledStartDate	datetime	x	Planned start date of the work order	
	ScheduledEndDate	datetime		ScheduledEndDate	datetime	x	Planned completion date of the work order	
	PlannedCost	money		PlannedCost	money	x	Estimated or planned cost for the work order	
	ActualStartDate	datetime		ActualStartDate	datetime	x	Actual start date	
	ActualEndDate	datetime		ActualEndDate	datetime	x	Actual end date	

Table 23. ETL Mapping Fact Tables (Source: Authors)

CHAPTER 4: DATA VISUALIZATION AND MULTI-DIMENSIONAL ANALYSIS

4.1. Data Cube

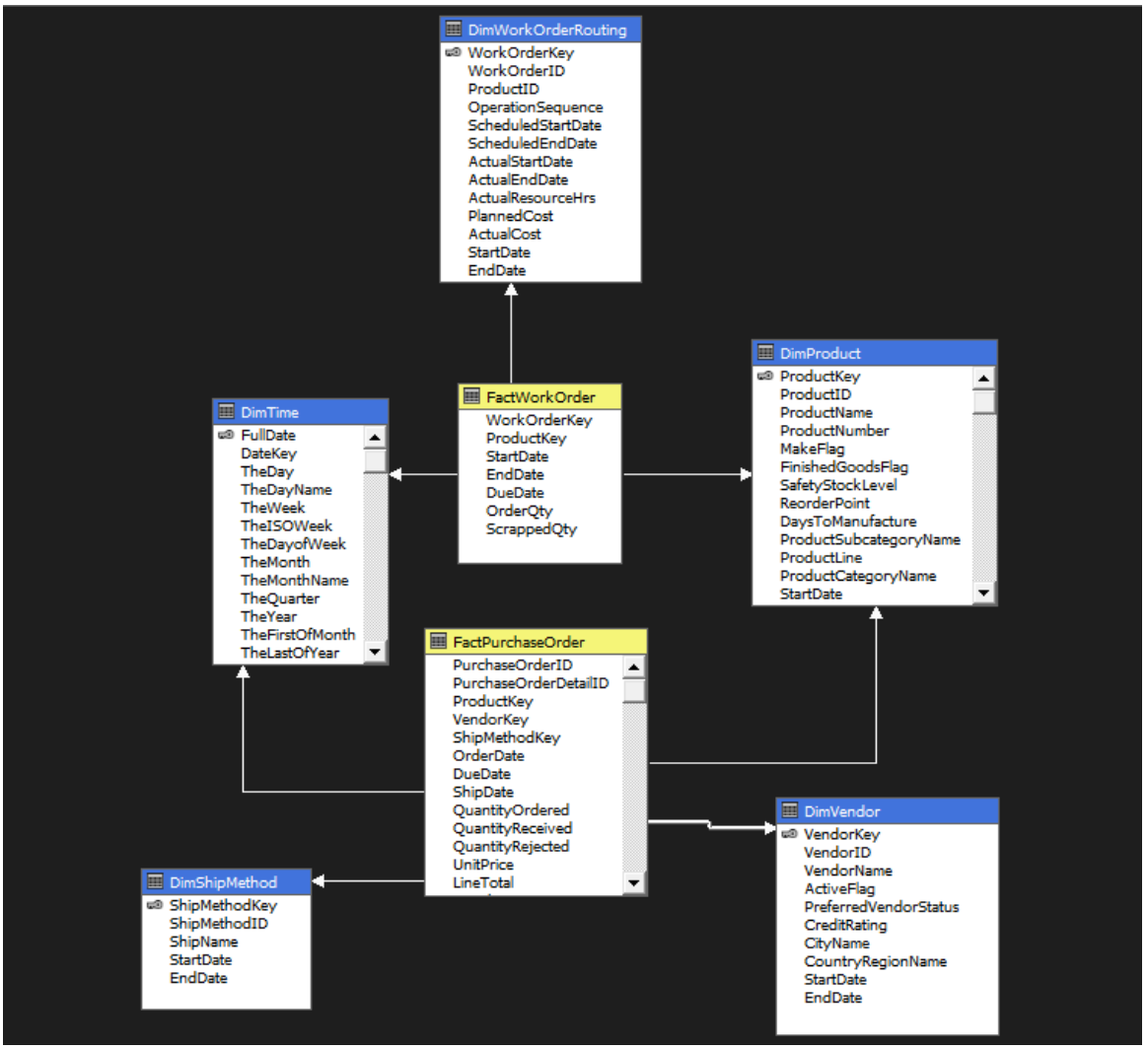


Figure 4.1. Data cube (Source: Authors)

This data cube represents a galaxy schema designed for analyzing manufacturing and procurement processes in an enterprise setting, particularly within the AdventureWorks data model. At the center is the FactWorkOrder table, which records manufacturing orders, including start and end dates, product keys, and quantities ordered or scrapped. Surrounding it are several dimension tables: DimProduct provides detailed information about the products, including their manufacturing characteristics; DimTime supports temporal analysis through a rich set of date-related

attributes; and DimWorkOrderRouting enables tracking of operations and resource planning tied to each work order.

In parallel, the FactPurchaseOrder fact table captures procurement activities, including purchase order details, vendor associations, unit prices, quantities, and dates. This table links to DimVendor—holding vendor profiles and metrics such as credit rating and active status—and DimShipMethod, which outlines the logistics options used in shipping. Together, this cube enables comprehensive analysis of the supply chain, vendor performance, product cost efficiency, and manufacturing timelines.

4.2. ETL Process

After multiple attempts and resolving numerous bugs, we developed a centralized control flow applicable to all Dimension and Fact tables.

This control flow is optimized for both logging efficiency and the implementation of Slowly Changing Dimensions (SCD) within the ETL process.

The workflow begins by capturing the user's IP address at the start of package execution. Since users may connect through different networks, the recorded IP addresses can vary. To prevent duplicate data during the staging phase, a `TRUNCATE` statement is executed. This removes all data from the staging tables without deleting the tables themselves. Once truncated, the data enters the staging phase, where it undergoes various sub-processes such as lookups and change detection. Based on the results, the data is routed to the appropriate staging tables and prepared for matching with the Dimension tables.

In the final step, the processed data is loaded into the Dimension tables.

4.2.1. DimProduct

- Control flow



Figure 4.2 . Control Flow (Source: Authors)

- ETL

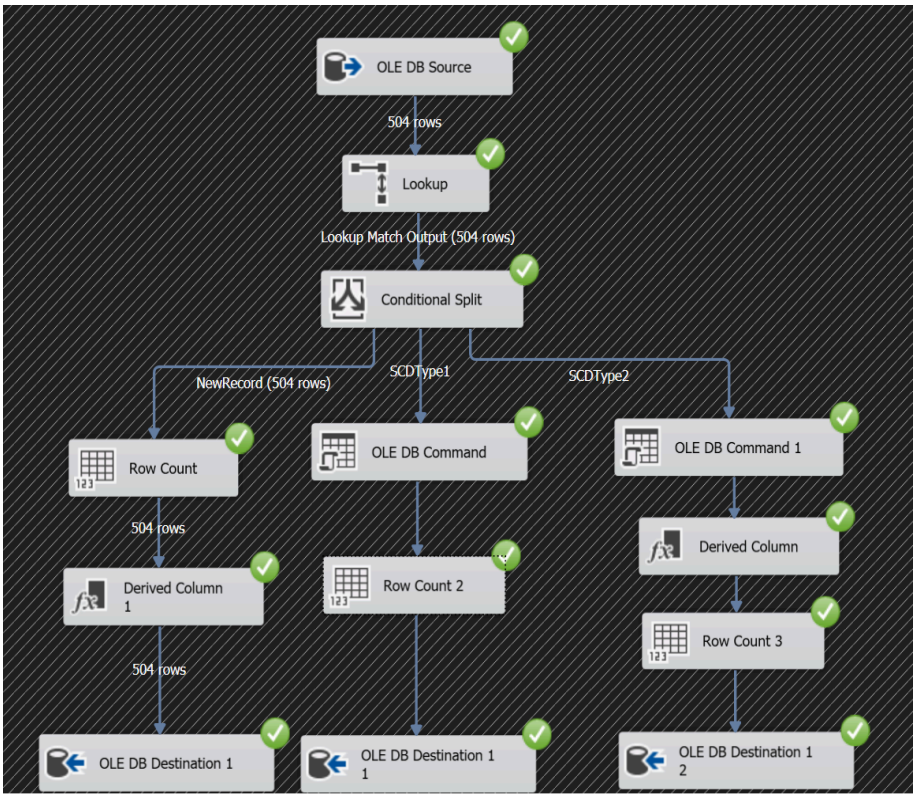


Figure 4.3. Data flow (Source: Authors)

4.2.2. DimShipMethod

- Control flow



Figure 4.4. Control Flow (Source: Authors)

- ETL

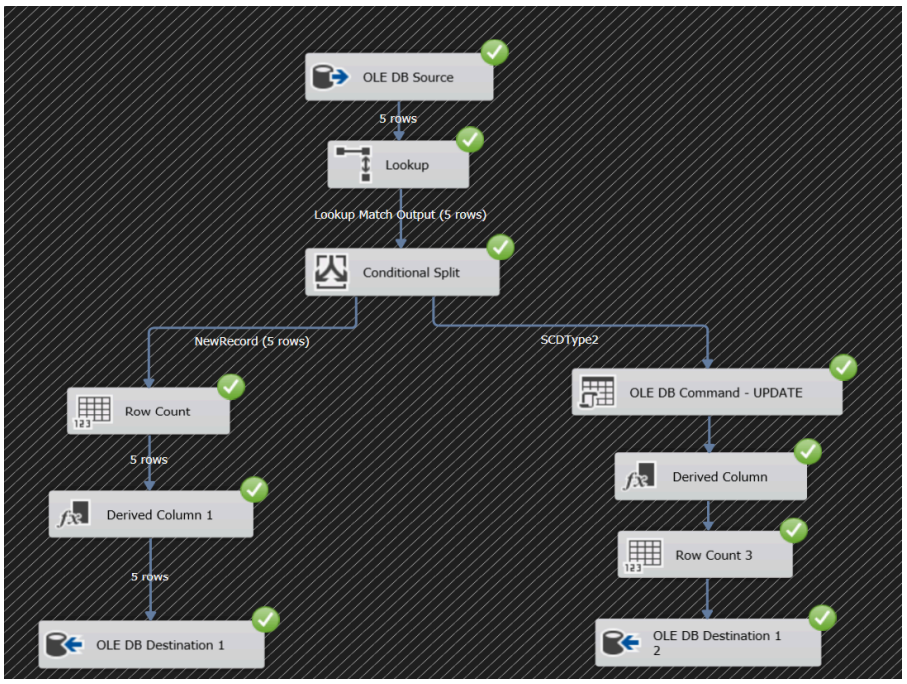


Figure 4.5 . Data flow (Source: Authors)

4.2.3. DimVendor

- Control Flow

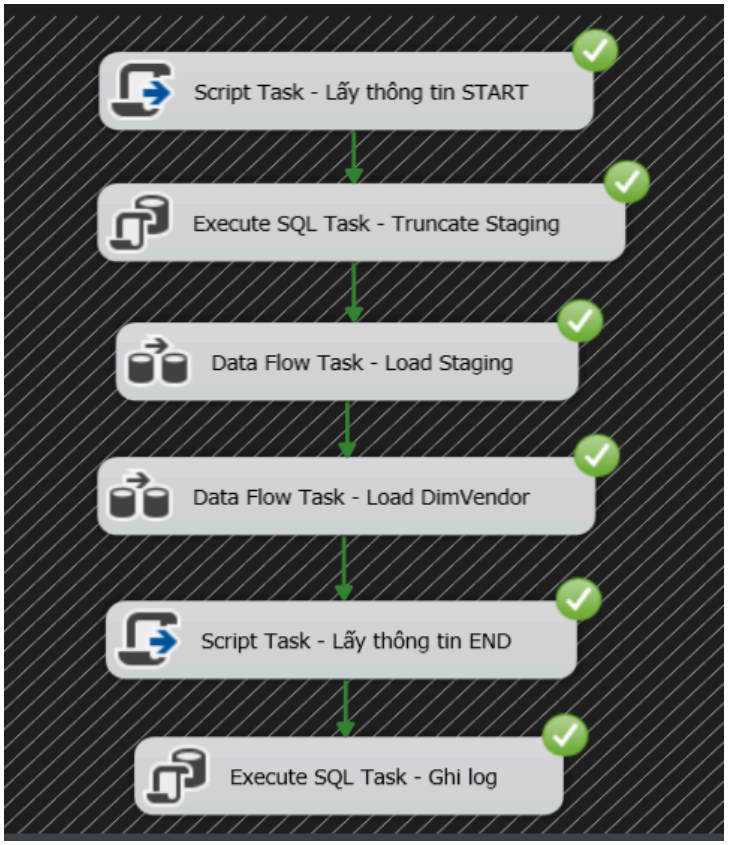


Figure 4.6 . Control Flow (Source: Authors)

- ETL

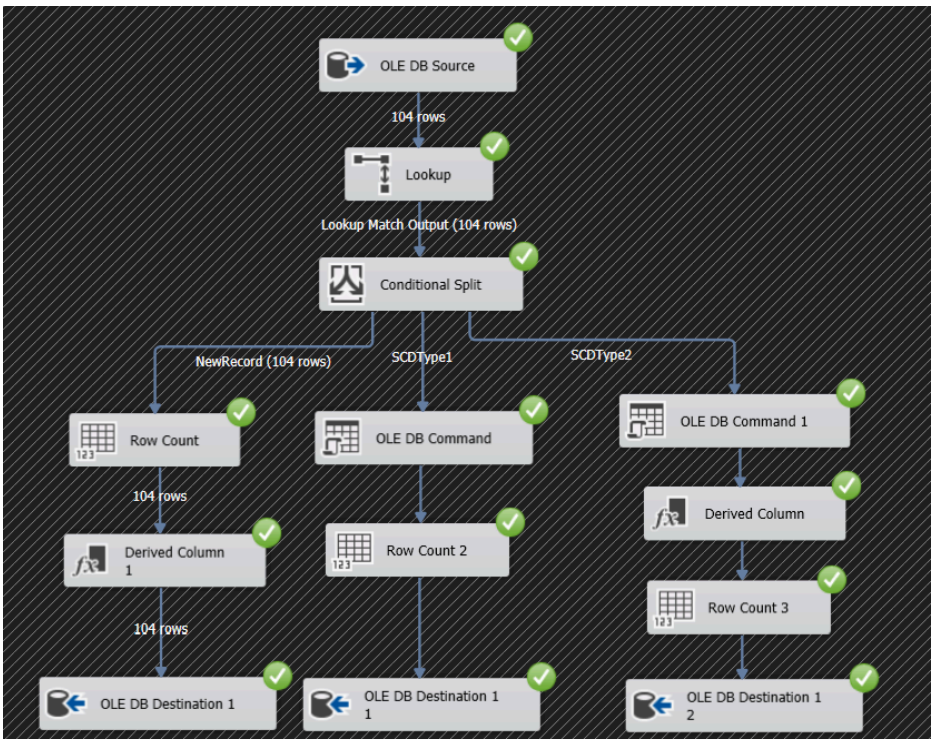


Figure 4.7 . Data Flow (Source: Authors)

4.2.4. DimScrapeSeason

- Control Flow



Figure 4.8 . Control Flow (Source: Authors)

- Data flow of DimScrapReason

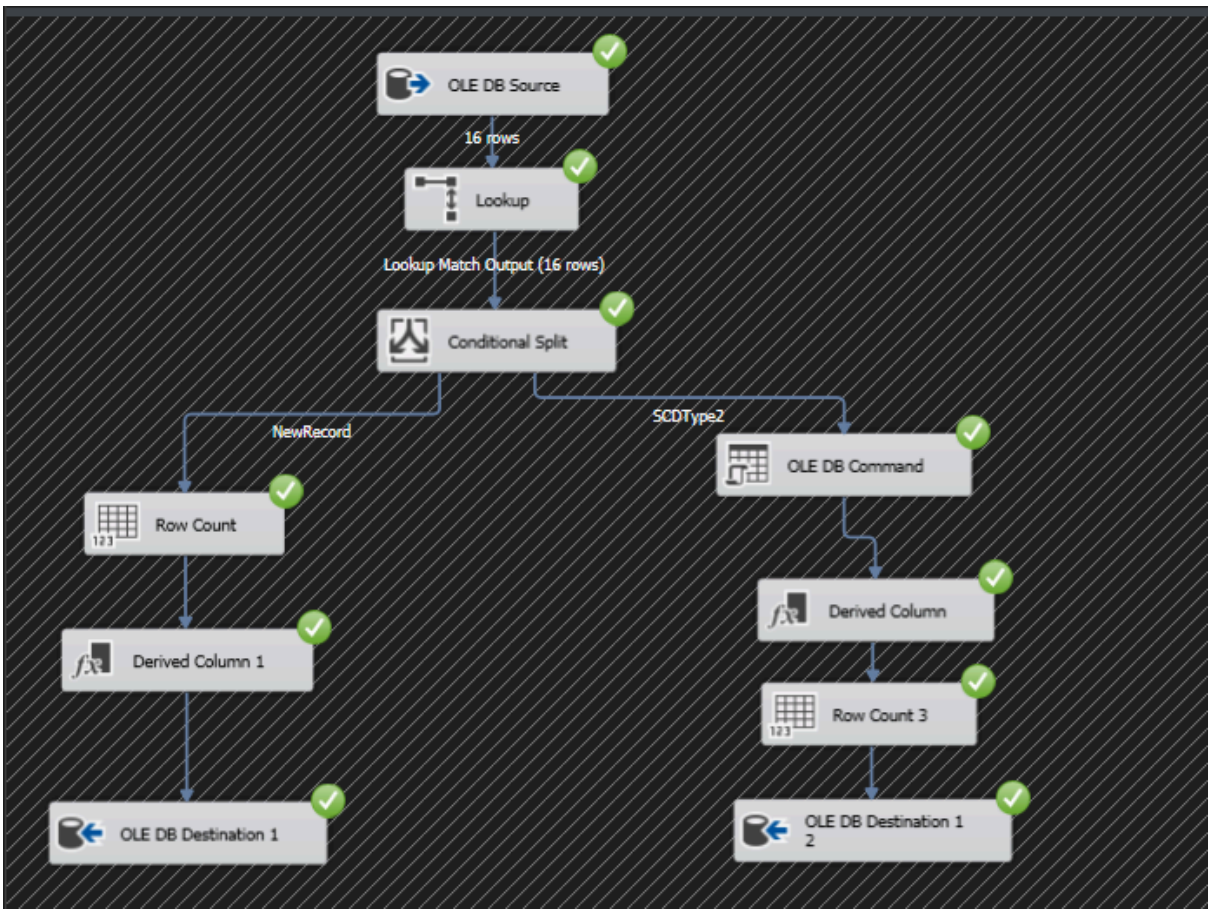


Figure 4.9 . Data Flow (Source: Authors)

4.2.5. FactPurchaseOrder

- Control Flow

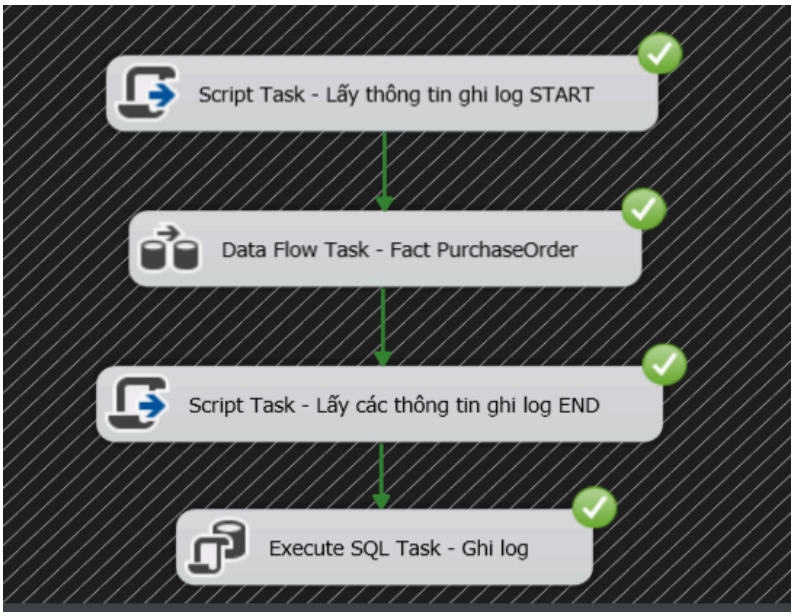


Figure 4.10 . Data flow (Source: Authors)

- ETL

Loading data on Fact Purchase Order

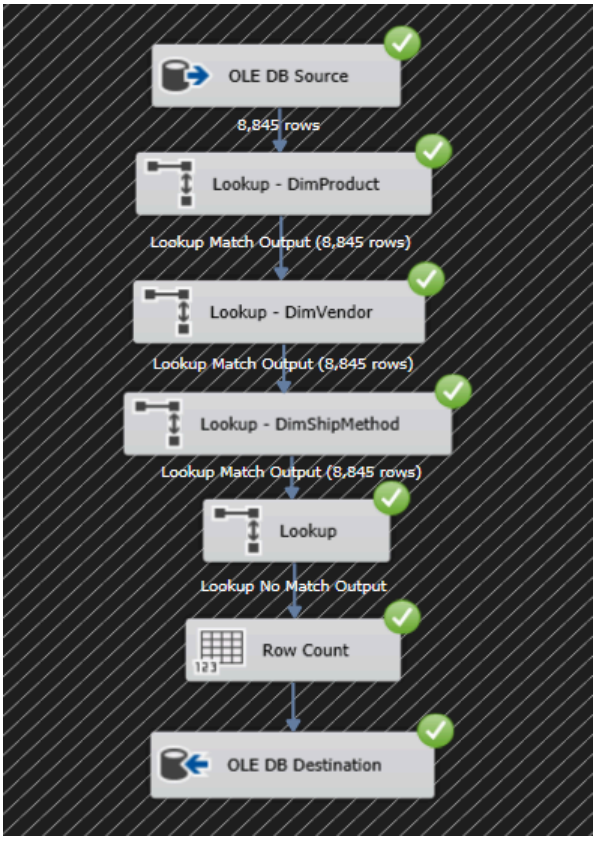


Figure 4.11. Data flow (Source: Authors)

4.2.6. FactWorkOrder

- Control flow

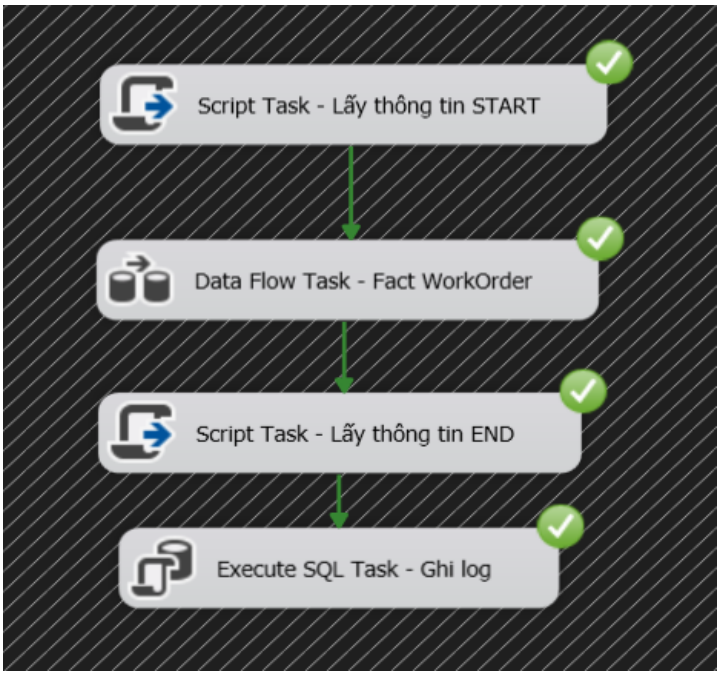


Figure 4.12. Control flow (Source: Authors)

- ETL

Loading data into FactWorkOrder

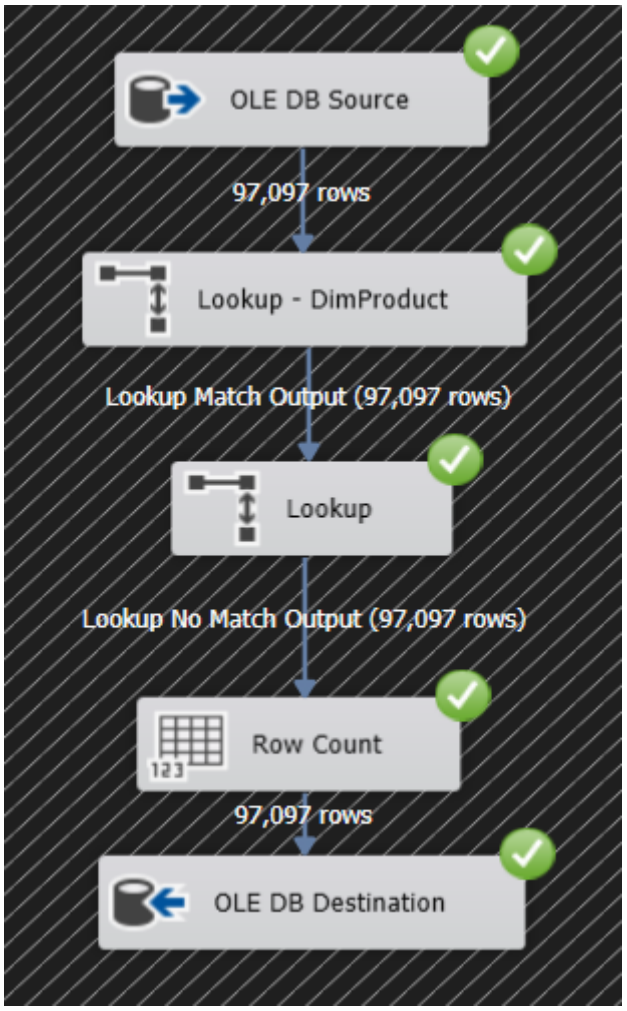


Figure 4.13. Data flow (Source: Authors)

4.2.7. DimTime

The SQL script creates and populates a DimTime table in the Galaxy database, designed to support time-based analysis in a Business Intelligence system. First, the Galaxy.DimTime table is created with a primary key named DateKey in the yyyymmdd format, along with various time-related attributes such as the day, day name, week number, month, quarter, year, and supporting columns like the first day of the month, last day of the year, and the day of the year.

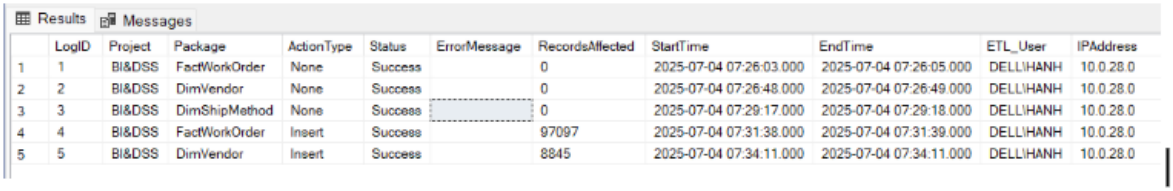
Next, two variables, @StartDate and @EndDate, are declared to define the time range for the data generation—in this case, from January 1, 2010, to December 31, 2015. A recursive Common Table Expression (CTE) named seq is used to generate a

continuous sequence of dates within this range. Then, a second CTE named src extracts various date attributes from each generated date, including day number, month name, ISO week number, quarter, first day of the month, last day of the year, and more.

Finally, the extracted data is inserted into the DimTime table using the INSERT INTO statement, with the MAXRECURSION 0 option to allow unlimited recursive iterations. The result is a complete, well-structured time dimension table, which can be used for slicing and filtering reports by day, month, quarter, or year in BI tools such as Power BI.

4.2.8. Log Table

We have developed an algorithm in C# to log every instance when a package is executed or activated. As part of this process, we capture the user's IP address at the time of activation. Since connecting to different networks may result in different IP addresses, we also record the associated username. This information is written to the database via a Script Task. By maintaining this log, we can effectively trace user activity and investigate any issues by reviewing the historical records.



Results | Messages

	LogID	Project	Package	ActionType	Status	ErrorMessage	RecordsAffected	StartTime	EndTime	ETL_User	IPAddress
1	1	BI&DSS	FactWorkOrder	None	Success		0	2025-07-04 07:26:03.000	2025-07-04 07:26:05.000	DELL\HANH	10.0.28.0
2	2	BI&DSS	DimVendor	None	Success		0	2025-07-04 07:26:48.000	2025-07-04 07:26:49.000	DELL\HANH	10.0.28.0
3	3	BI&DSS	DimShipMethod	None	Success		0	2025-07-04 07:29:17.000	2025-07-04 07:29:18.000	DELL\HANH	10.0.28.0
4	4	BI&DSS	FactWorkOrder	Insert	Success		97097	2025-07-04 07:31:38.000	2025-07-04 07:31:39.000	DELL\HANH	10.0.28.0
5	5	BI&DSS	DimVendor	Insert	Success		8845	2025-07-04 07:34:11.000	2025-07-04 07:34:11.000	DELL\HANH	10.0.28.0

4.3. Variables

We defined variables to support the ETL process, each with its own specific purpose.

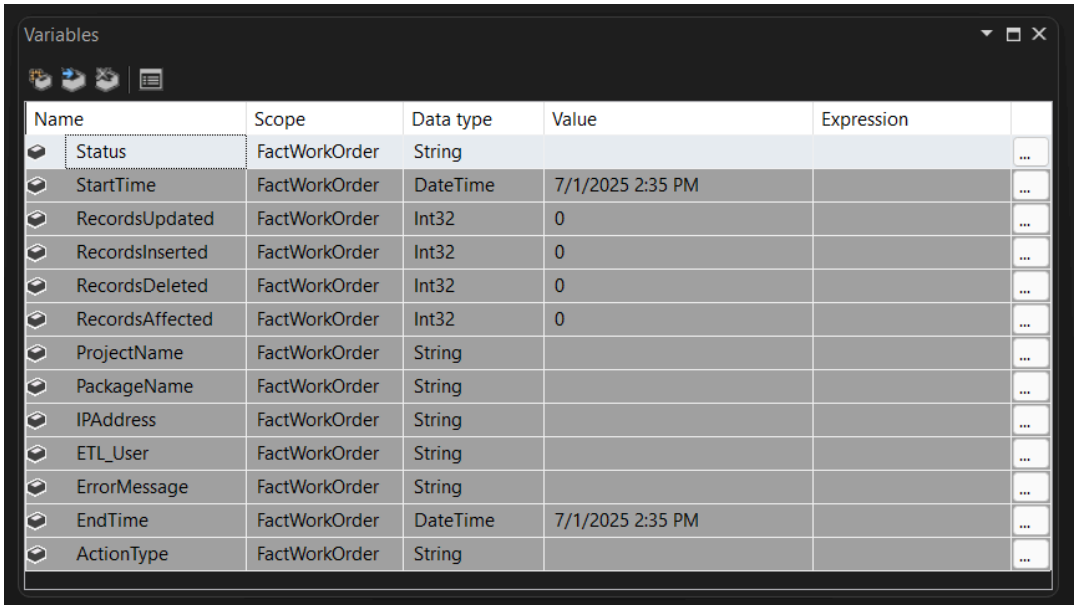


Figure 4.14 . Variables (Source: Authors)

4.4. Data Pipeline

After completing the ETL (Extract, Transform, Load) process, the result is a structured data pipeline that automates the flow of clean, integrated, and reliable data from source systems to our target storage—such as a data warehouse or analytics platform. This pipeline serves as the foundation for downstream activities like reporting, data visualization, and advanced analytics, ensuring that decision-makers have timely access to high-quality data.

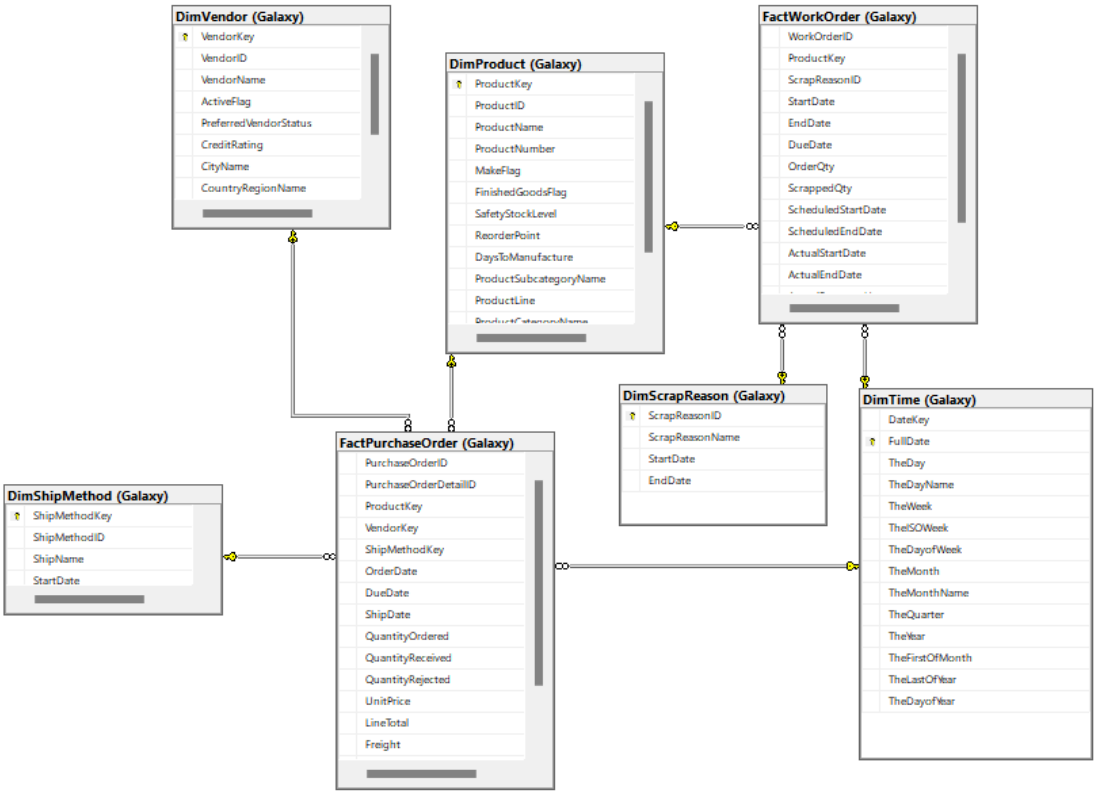


Figure 4.15. Data Pipeline after ETL (Source by Authors)

4.5. Visualization

4.5.1. Overview

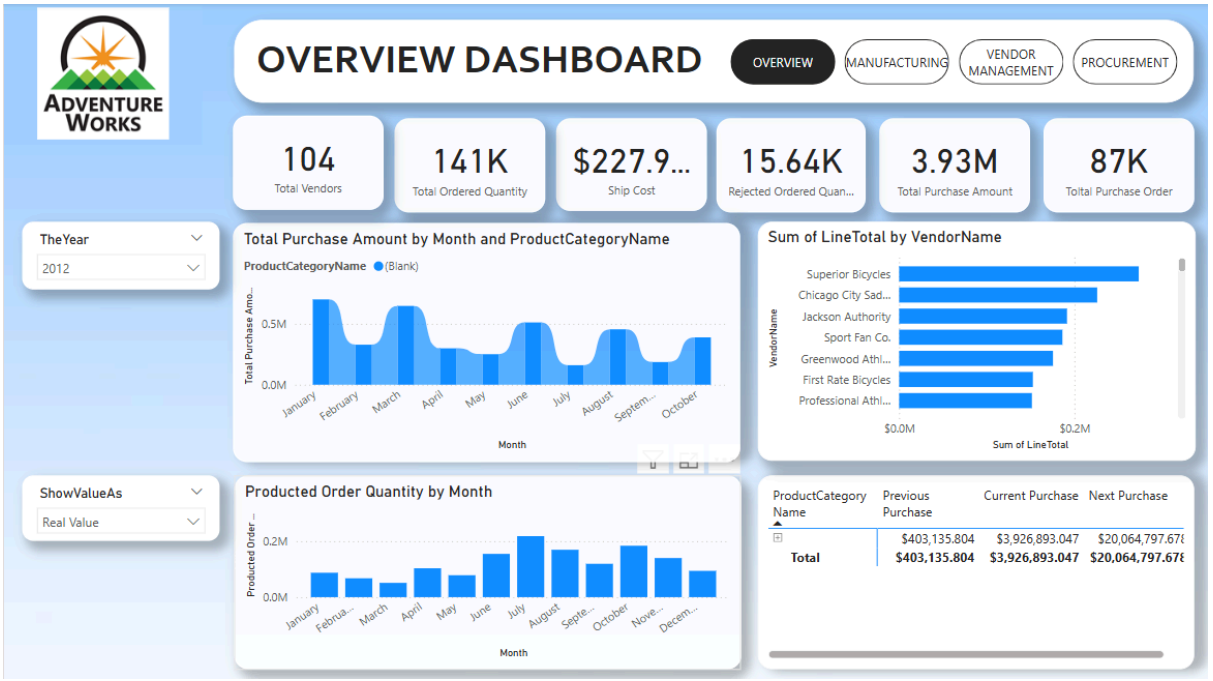


Figure 4.16. Overview Dashboard (Source: Authors)

The Adventure Works Overview Dashboard presents a clear and comprehensive summary of the company's procurement performance in 2012. At the top, six key indicator cards highlight essential metrics: the company worked with 104 vendors, placed orders totaling 141K items, and incurred approximately $227.9K in shipping costs. Additionally, there were 15.64K rejected items, with a total purchase value of $3.93 million and 87K purchase orders issued throughout the year.

The dashboard includes multiple visualizations to support data-driven decisions. A bar chart labeled "Total Purchase Amount by Month" displays how monthly purchasing trends evolved over the year, though the product category data appears to be missing. Another chart, "Produced Order Quantity by Month", shows a clear spike in July, indicating a seasonal increase in production activity. A horizontal bar chart titled "Sum of LineTotal by VendorName" identifies the top vendors based on purchase value, where Superior Bicycles and Chicago City Saddles stand out as the highest contributors. Furthermore, a detailed table compares previous, current, and projected future purchases, providing insight into the company's procurement growth, with forecasts reaching over $20 million.

Interactive filters such as Year and ShowValueAs allow users to adjust the dashboard view based on selected periods and value formats. Overall, this dashboard offers a high-level yet insightful look into purchasing operations, equipping managers and procurement teams with the necessary information to evaluate performance, manage vendors effectively, and make informed strategic decisions.

4.5.2. Manufacturing



Figure 4.17. Manufacturing Dashboard (Source: Authors)

The Manufacturing Dashboard offers a comprehensive overview of production performance within the AdventureWorks company, enabling stakeholders to monitor key manufacturing metrics in real time. With essential KPIs such as total ordered quantity, actual production cost, total resource hours, productivity per resource hour, and average production duration, users gain immediate insight into operational efficiency. The dashboard highlights scrap quantity by product, identifying components like BB Ball Bearing and Seat Stays as significant contributors to production waste. A monthly waste cost line chart reveals cost fluctuations, with notable spikes in January and August, suggesting periods of inefficiency that warrant further investigation. Additionally, the dashboard visualizes the relationship between total order quantity and productivity over the year, helping managers track seasonal trends. A detailed table breaks down order volume, scrap quantity, production duration, waste cost, and unit cost per product, offering granular insights that support cost control and process optimization. Altogether, the dashboard serves as a vital tool for improving manufacturing outcomes and reducing waste through informed, data-driven decisions.

Through interactive visuals and performance metrics, this dashboard empowers stakeholders to pinpoint inefficiencies, minimize waste, and enhance overall manufacturing effectiveness.

4.5.3. Vendor Management

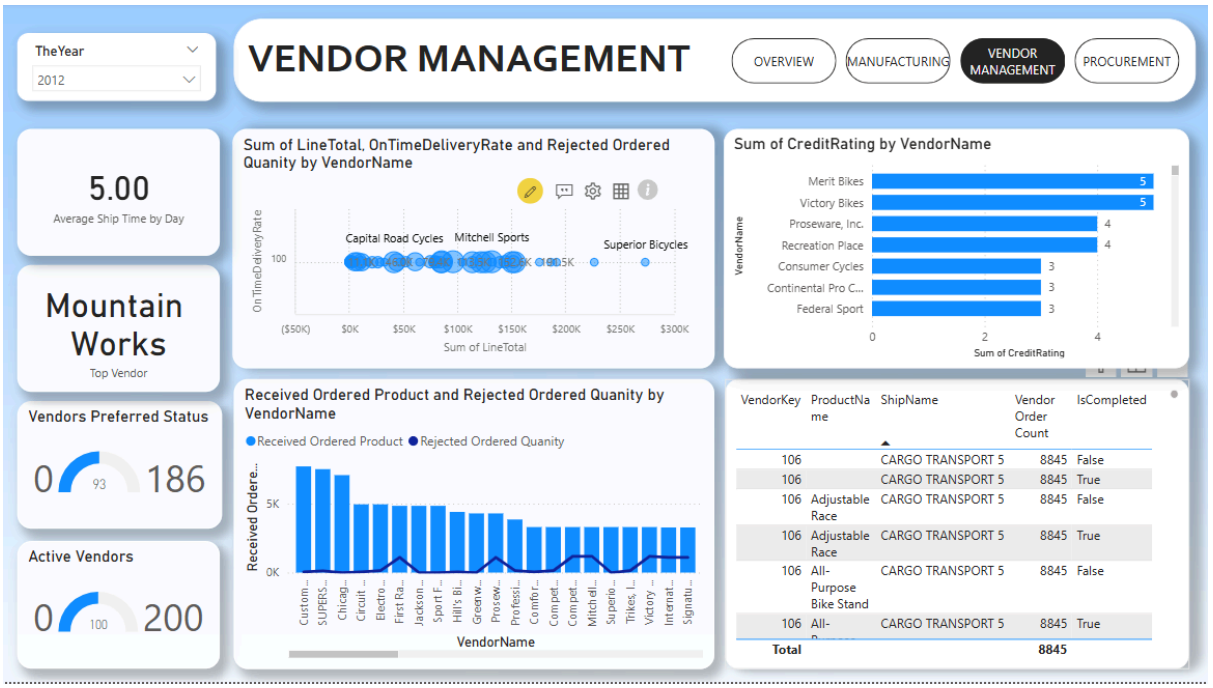


Figure 4.18. Vendor Management Dashboard (Source: Authors)

The Vendor Management Dashboard provides a strategic overview of supplier performance and vendor-related activities, enabling the Purchasing department to evaluate and optimize supplier relationships. Key performance indicators such as average ship time (5 days), top vendor (Mountain Works), and the number of active vendors (200) are clearly displayed to offer immediate context. The scatter plot visualizes vendor performance by correlating total spending (LineTotal) with on-time delivery rate and rejected order quantity, helping identify reliable suppliers such as Capital Road Cycles and Mitchell Sports. A bar chart of credit ratings by vendor reveals the top-rated suppliers, including Merit Bikes and Victory Bikes, each with a rating of 5. Additionally, the dashboard compares received vs. rejected order quantities across vendors, offering insight into supply consistency and quality issues. Complementing this, a detailed vendor order table displays shipping information, order counts, and completion status, allowing for in-depth performance tracking.

Altogether, this dashboard equips stakeholders with the tools needed to assess vendor reliability, streamline procurement operations, and strengthen supply chain efficiency.

4.5.4. Procurement

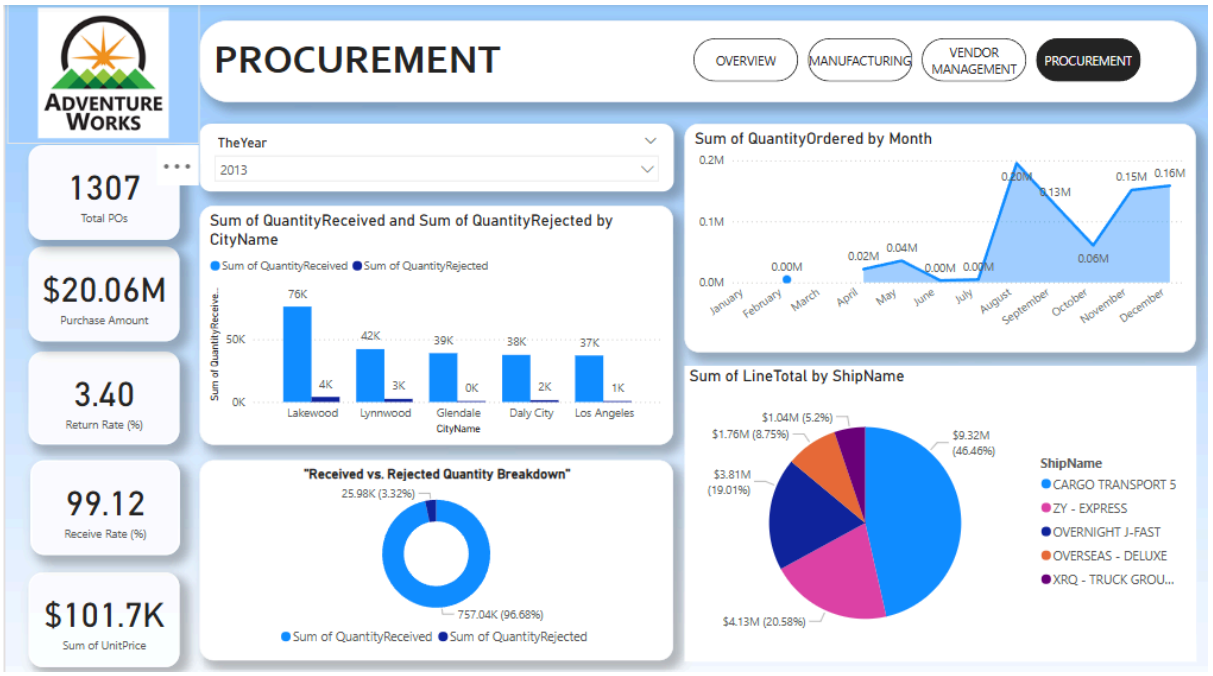


Figure 4.19. Procurement Dashboard (Source: Authors)

The Procurement Dashboard for Adventure Works offers a comprehensive overview of the company's purchasing operations, encompassing total purchase orders, financial values, delivery quality, and the performance of logistics partners.

Positioned on the left side of the dashboard, key performance indicators (KPIs) serve as immediate reference points for overall operational efficiency. In 2013, the company processed 1,307 purchase orders, amounting to a total procurement value exceeding $20 million USD. These high-level metrics form the basis for deeper analysis of material flows within the organization. A return rate of 3.4% indicates a relatively small proportion of goods failing to meet quality or compliance standards, whereas a receive rate of 99.12% suggests a high degree of reliability in supplier fulfillment.

Central to the dashboard are various visualizations that enable multidimensional analysis of procurement data. The clustered column chart displaying quantities received and rejected by city provides geographic insight into logistics performance. Cities such as Lakewood demonstrate a significantly higher volume and more stable

receiving process, while other locations manage smaller volumes and exhibit varying levels of delivery consistency—potentially signaling localized operational issues.

Meanwhile, the line chart illustrating monthly ordering trends reveals seasonal procurement behavior. The surge in purchasing activity observed in August and September may reflect efforts to meet year-end demand or to prepare for cyclical production requirements. Such temporal patterns are invaluable for inventory forecasting and procurement planning.

In terms of quality control, the donut chart visually contrasts the proportion of accepted and rejected quantities. Although the vast majority of goods were received without issue, the small percentage of rejections warrants closer monitoring of quality assurance procedures and supplier selection criteria.

Finally, the pie chart analyzing cost distribution by shipping provider uncovers a concentrated reliance on a limited number of logistics partners. One shipping company alone accounts for nearly half of total transportation expenditure. While this may reflect a high degree of operational trust, it simultaneously introduces a potential risk of overdependence, emphasizing the need for risk mitigation strategies.

In sum, this dashboard serves not only as a real-time monitoring tool for daily operations but also as a strategic foundation for decision-making in supply chain optimization and procurement cost control. The integration of quantitative metrics with clear data visualizations facilitates data-driven insights, enabling managers to initiate timely and evidence-based improvements aligned with broader organizational objectives.

CHAPTER 5: CONCLUSIONS AND FUTURE WORKS

5.1. Achieved Results

Following the research and implementation phase, the Business Intelligence (BI) project yielded several notable outcomes. Initially, the project team identified key business requirements and key performance indicators (KPIs), alongside conducting thorough analysis and documentation of relevant data sources. Based on these findings, a comprehensive and well-defined KPI framework was developed, aligned with the project's objectives. This was accompanied by the construction of both a centralized Data Warehouse and targeted Data Marts, serving as foundational components for effective data integration and analysis.

Data integration was accomplished using ETL and ELT methodologies, with the aid of SQL Server Integration Services (SSIS) to extract, transform, and load data from various heterogeneous sources in a reliable and efficient manner. This established a solid basis for multidimensional data analysis and enabled flexible and accurate KPI deployment through SQL Server Analysis Services (SSAS).

Finally, the reporting and dashboard system was implemented using Power BI, providing a dynamic and user-friendly interface. These visualizations facilitated data exploration and interpretation, thereby empowering users to access actionable insights and make informed decisions based on processed analytical data.

5.2. Future development directions

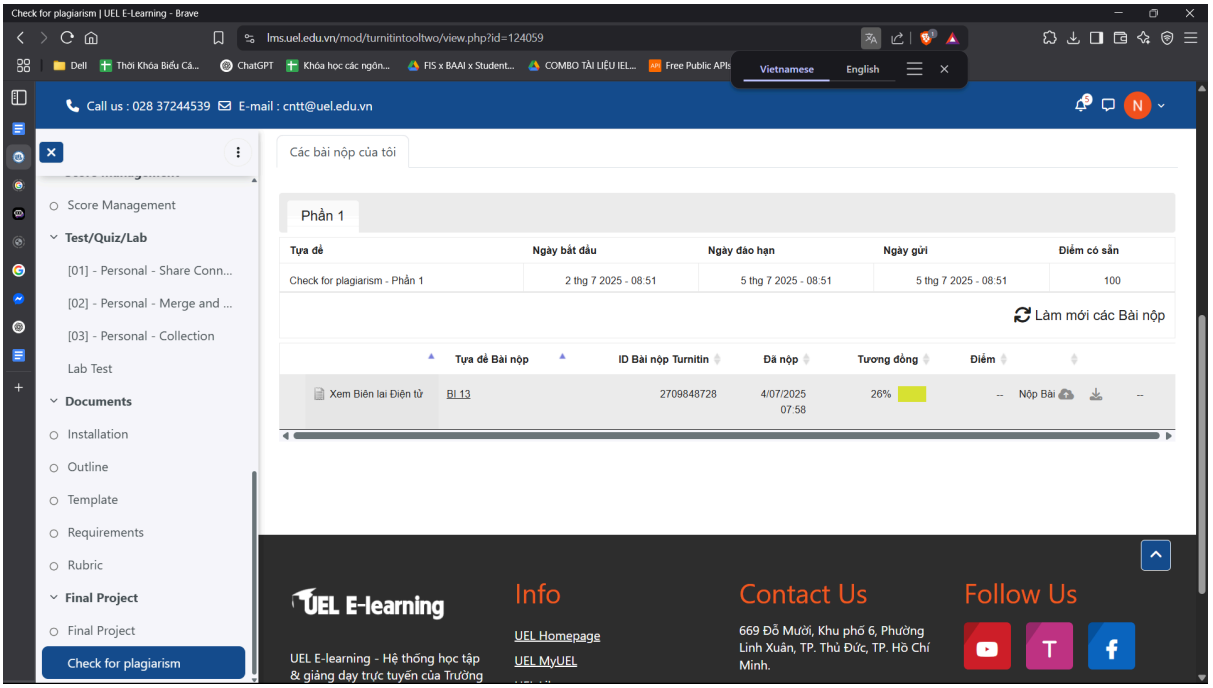
To enhance the effectiveness and overall value of the Business Intelligence (BI) system in subsequent phases, several improvement strategies should be considered. First and foremost, it is essential to strengthen data quality control mechanisms. This includes implementing more rigorous monitoring procedures to proactively identify issues such as missing values (null), data inconsistencies, and invalid entries, thereby ensuring the accuracy and reliability of the entire analytical infrastructure.

In addition, resolving discrepancies between data tables—particularly those involving the Product and Inventory tables—should be prioritized to improve the completeness and consistency of the dataset. Cross-referencing techniques and regular data updates are recommended to enhance the quality and trustworthiness of analytical reports.

Another critical direction is the optimization of the multidimensional data cube design process, which would contribute to improved data retrieval performance and analytical efficiency. This involves standardizing cube construction methodologies and maintaining clear documentation to reduce ambiguity and support smooth operational workflows.

Furthermore, integrating advanced analytics techniques—such as predictive modeling and trend analysis—into the BI ecosystem would significantly expand forecasting capabilities, thereby supporting more informed strategic decision-making. Simultaneously, it is crucial to invest in continuous end-user training, provide ongoing technical support, and establish feedback mechanisms to refine the system. These efforts will ensure that the BI platform remains aligned with evolving business needs and continues to deliver meaningful organizational value.

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